

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

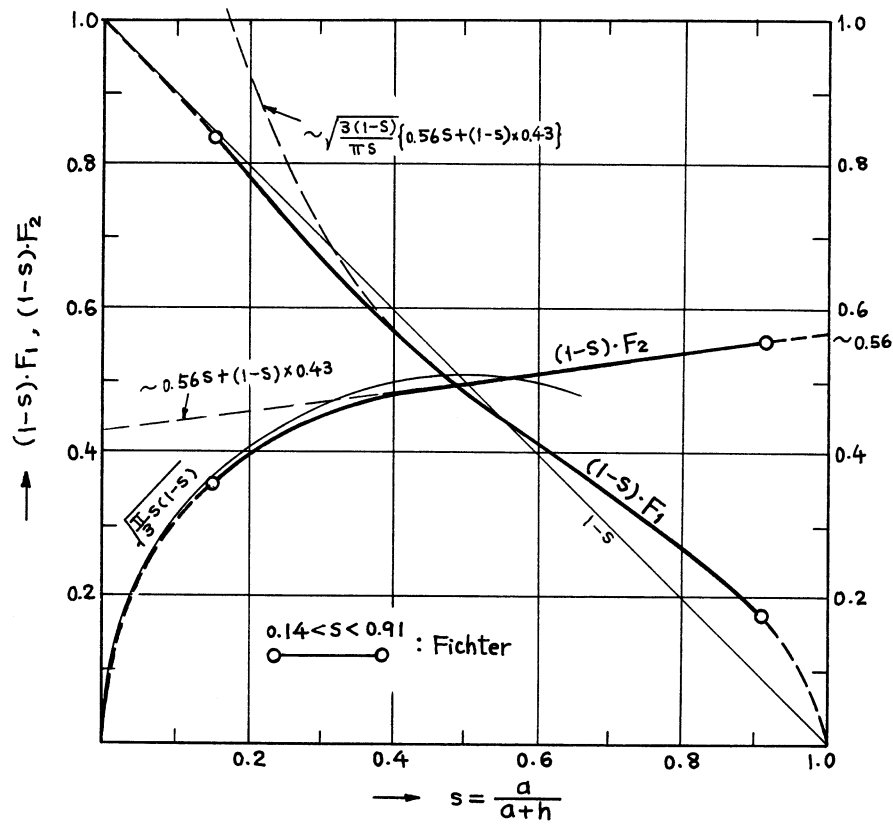
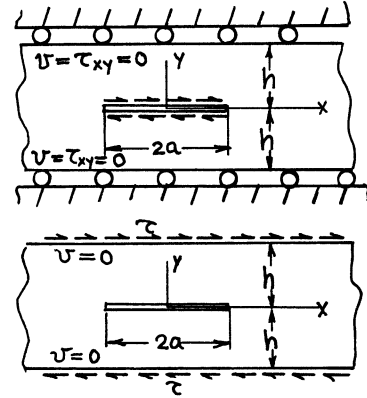
$$\begin{aligned} K_{II} &= \tau \sqrt{\pi a} \cdot F_1(s) \\ &= \tau \sqrt{3h} \cdot F_2(s) \end{aligned}$$

$$s \rightarrow 0 \left(\frac{a}{h} \rightarrow 0 \right) : F_1 = 1, F_2 \rightarrow \sqrt{\frac{\pi a}{3h}}$$

$$s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty \right) :$$

$$(1-s)F_2 \simeq 0.56s + (1-s) \times 0.43$$

$$(1-s)F_1 \simeq \sqrt{\frac{3h}{\pi a}} \{0.56s + (1-s) \times 0.43\}$$



Method: Fourier Transform

Accuracy: Order of 1%

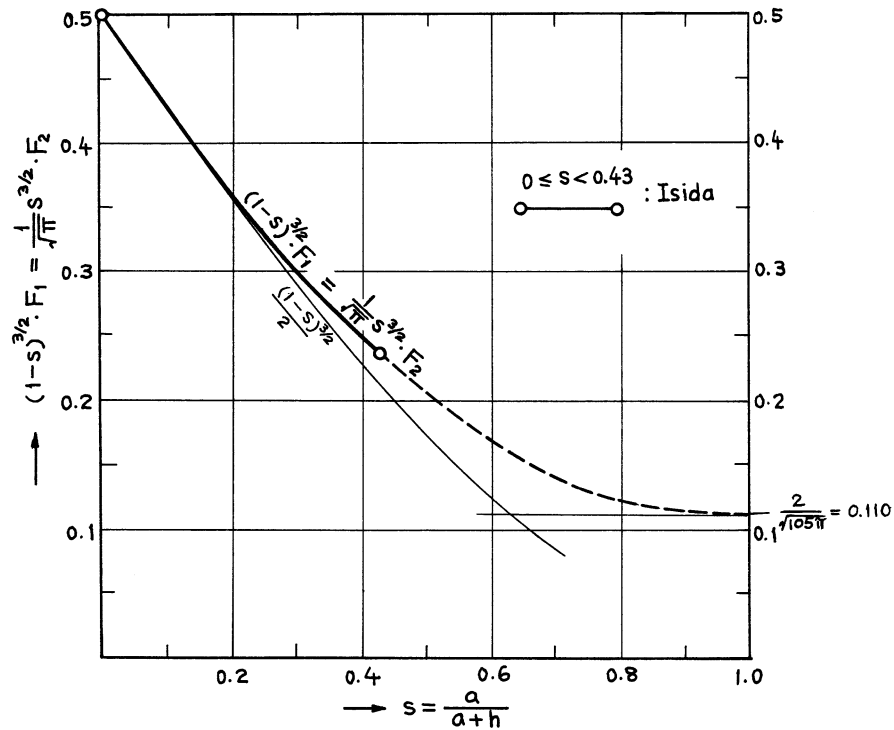
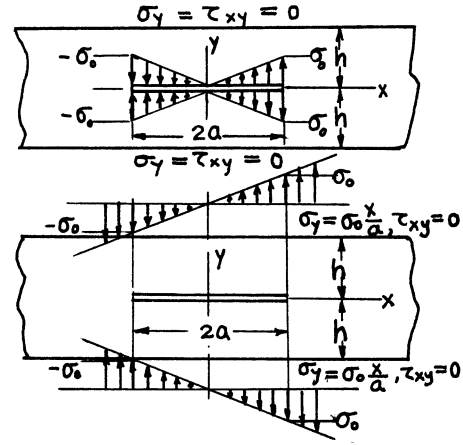
Reference: **Fichter 1967**

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$\begin{aligned} K_I &= \sigma_0 \sqrt{\pi a} \cdot F_1(s) \\ &= \sigma_0 \sqrt{h} \left(\frac{a}{h}\right)^2 \cdot F_2(s) \end{aligned}$$

$$\begin{aligned} s \rightarrow 0 \left(\frac{a}{h} \rightarrow 0\right) : F_1 &= \frac{1}{2} \\ F_2 &\rightarrow \frac{\sqrt{\pi}}{2} \left(\frac{h}{a}\right)^{3/2} \end{aligned}$$

$$\begin{aligned} s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty\right) : F_2 &\simeq \frac{2}{\sqrt{105}} \\ F_1 &\simeq \frac{2}{\sqrt{105\pi}} \left(\frac{a}{h}\right)^{3/2} \end{aligned}$$



Methods: Expansions of Complex Stress Potentials (Isida, $0 < s < 0.43$, Interpolated Asymptotically (Tada $0.43 < s < 1$), $s \rightarrow 1$ Theory of Beam Bending

Accuracy: 1% for $0 < s < 0.43$, better than 5% for $0.43 < s < 1$ (exact for $s \rightarrow 0$ and $s \rightarrow 1$)

References: **Isida 1971a; Tada 1973**

$$s = \frac{a}{a+h} = \frac{a/h}{1+a/h} = \frac{1}{1+h/a}$$

$$K_I = \sigma_0 \sqrt{\pi a} \cdot F_1(s)$$

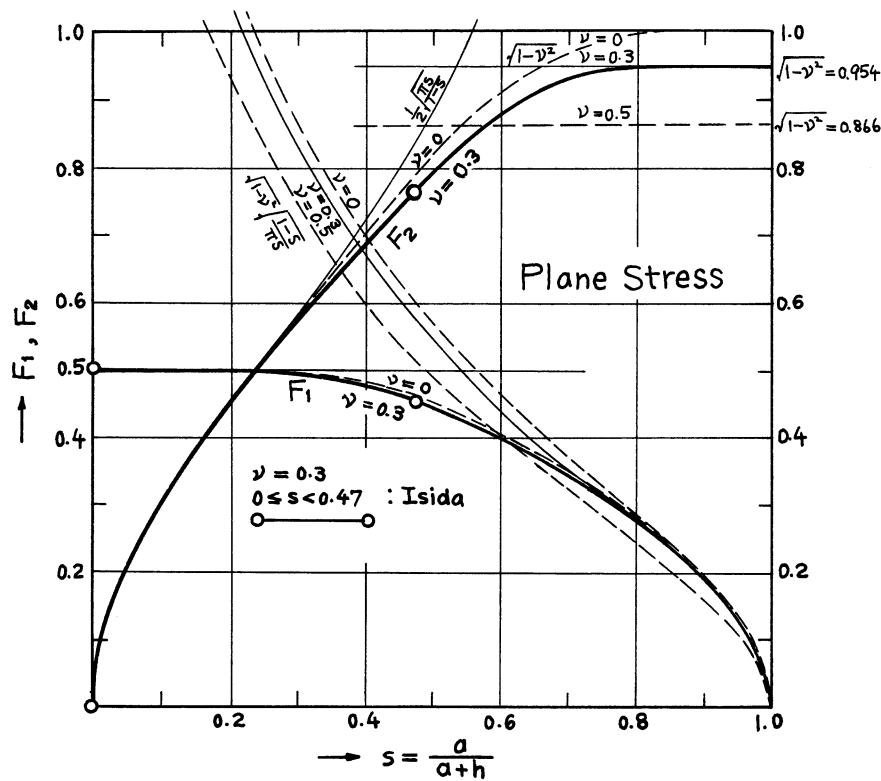
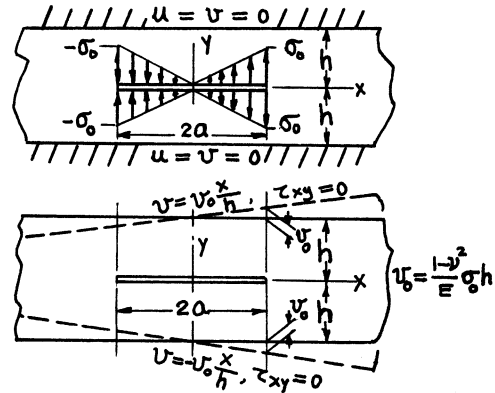
$$= \sigma_0 \sqrt{h} \cdot F_2(s)$$

$$s \rightarrow 0 \left(\frac{a}{h} \rightarrow 0 \right) : F_1 = \frac{1}{2}$$

$$F_2 \rightarrow \frac{1}{2} \sqrt{\frac{\pi a}{h}}$$

$$s \rightarrow 1 \left(\frac{a}{h} \rightarrow \infty \right) : F_2 = \sqrt{1-\nu^2}$$

$$F_1 \rightarrow \sqrt{1-\nu^2} \sqrt{\frac{h}{\pi a}}$$

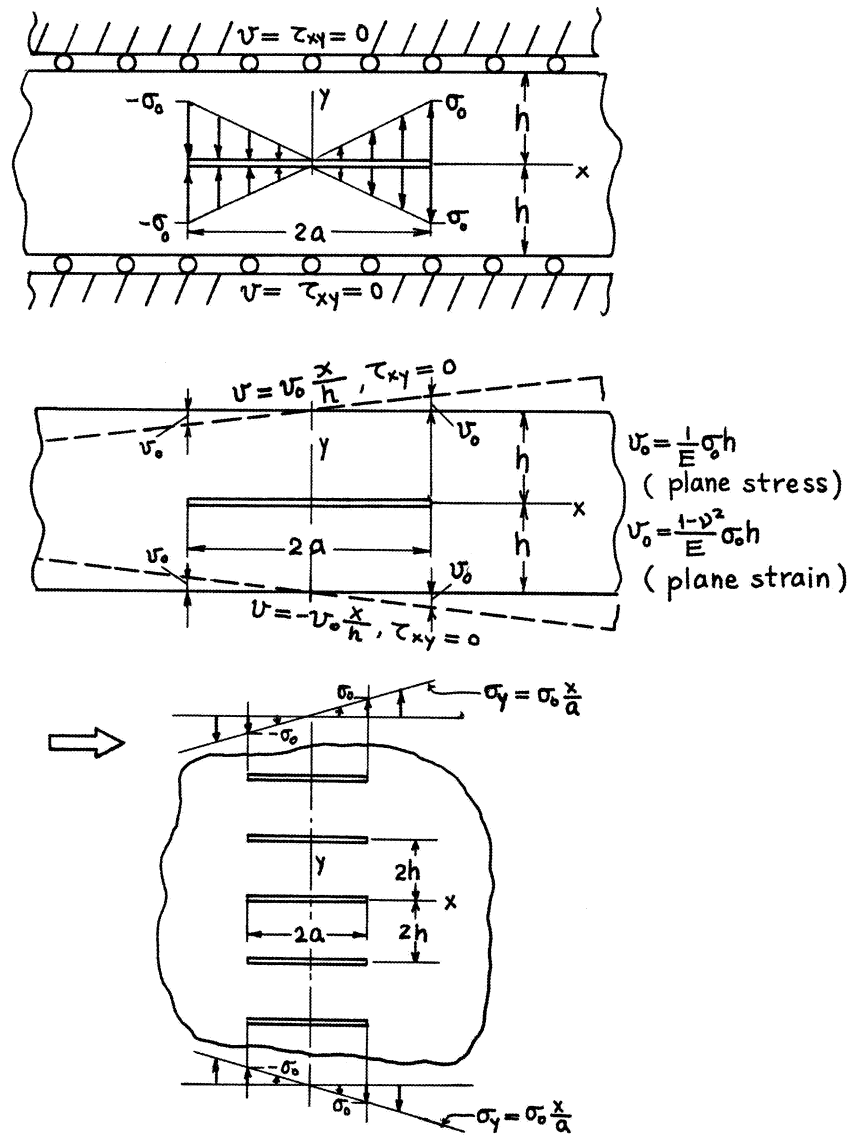


Methods: Expansions of Complex Stress Potentials (Isida, $s < 0.47$), Interpolated Asymptotically (Tada, $0.47 \leq s < 1$); $s \rightarrow 1$: Energy Balance (Rice)

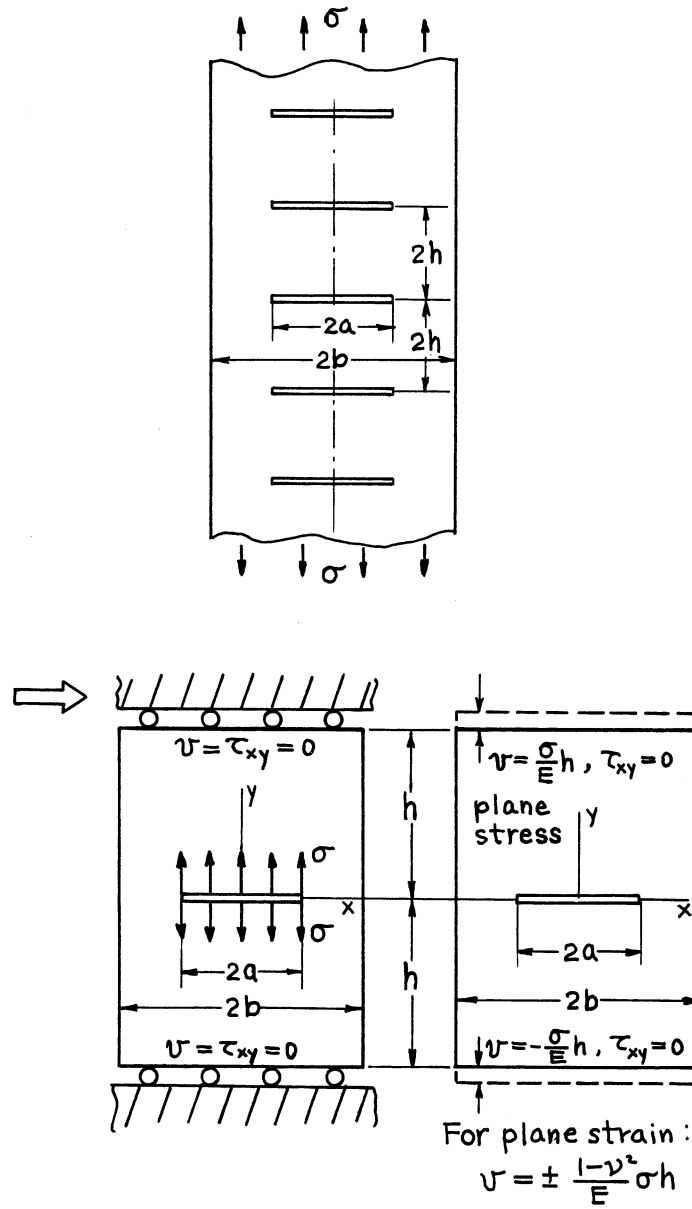
Accuracy: 1% for $s < 0.47$, estimated at better than 3% for $0.47 < s < 1$ (exact for $s \rightarrow 0$ and $s \rightarrow 1$)

References: **Rice 1967; Isida 1971a; Tada 1973**

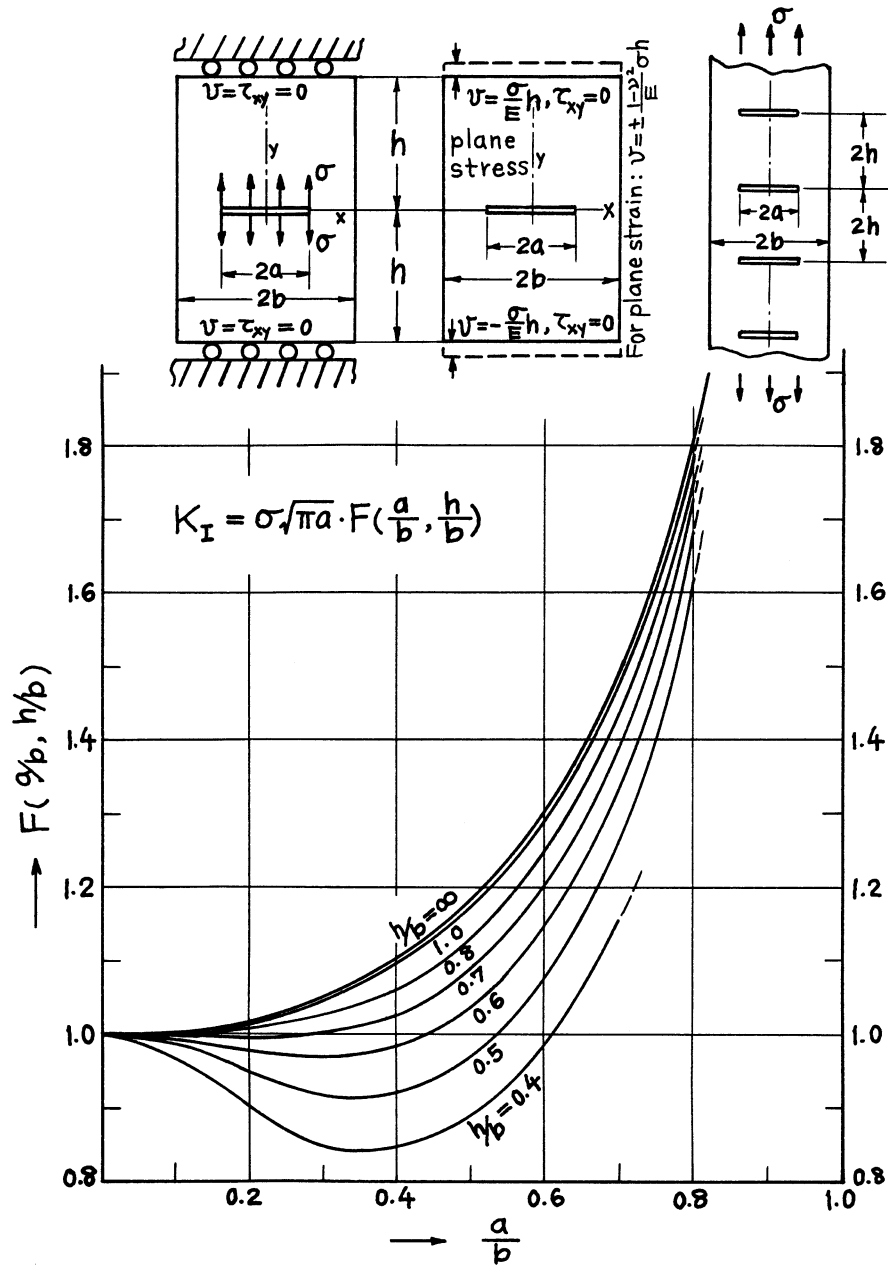
NOTE: For plane strain, $F_2(s \rightarrow 1) = \frac{\sqrt{1-2\nu}}{1-\nu}$, etc.; for $\nu = 0$ see page 14.4.



See page 14.4.



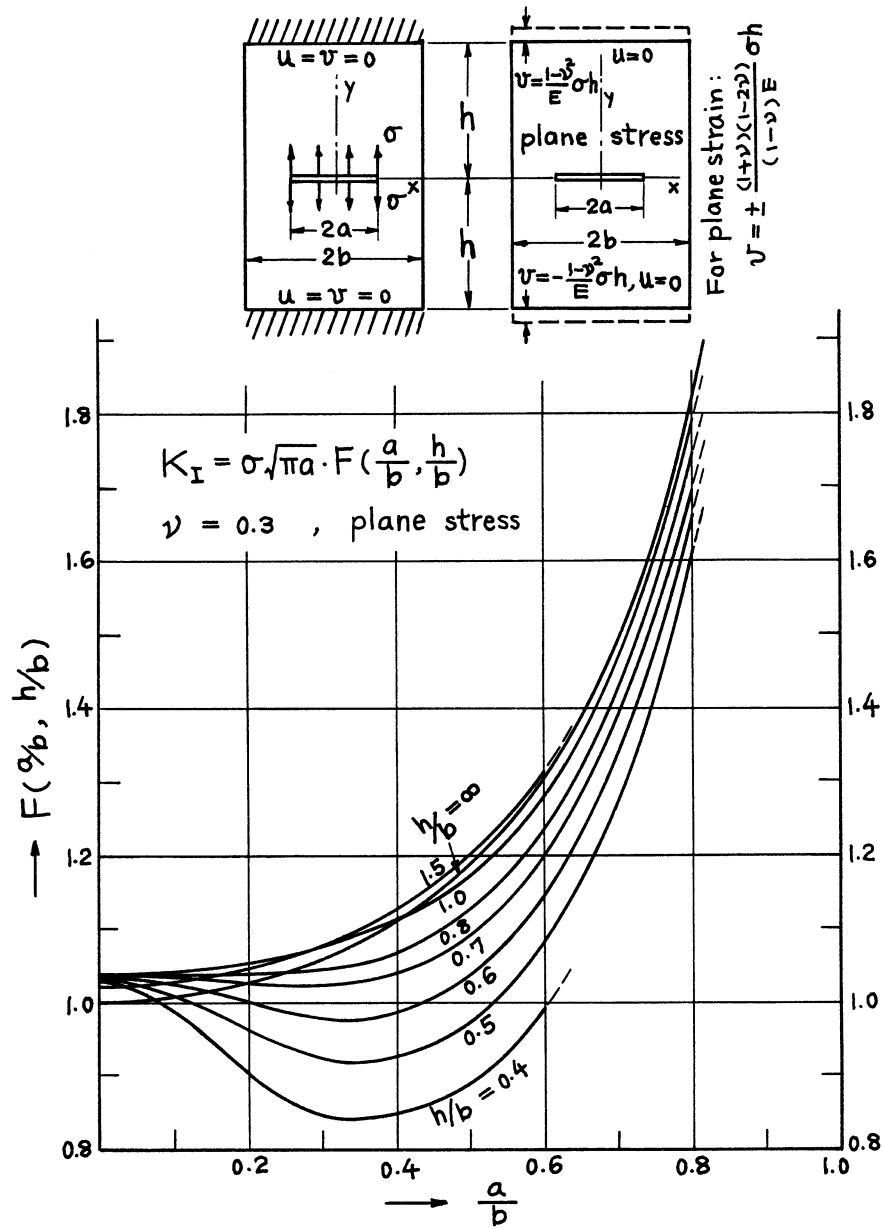
See page 18.2.



Method: Laurent's Expansion of Complex Stress Potentials

Accuracy: Better than 1%

Reference: **Isida 1971a,b**



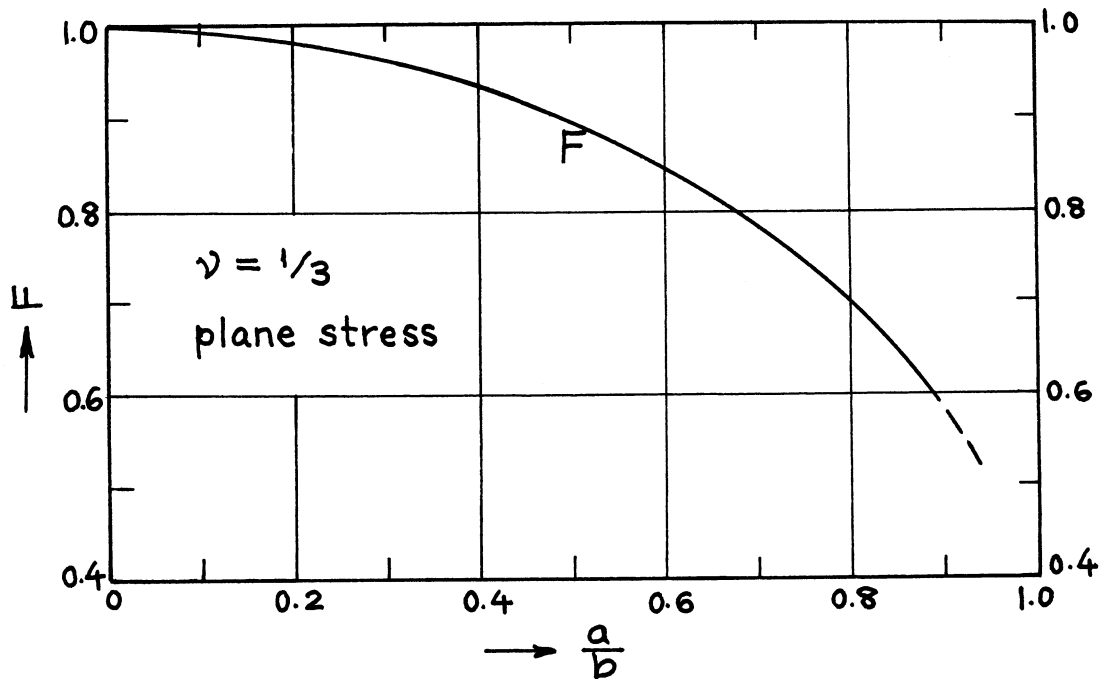
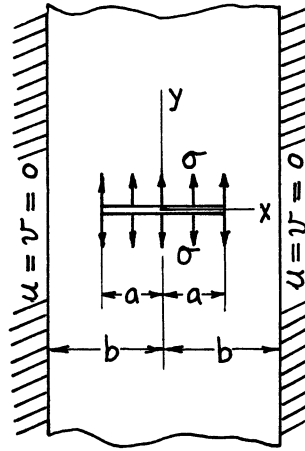
Method: Laurent's Expansion of Complex Stress Potentials Combined with a Boundary Collocation Method

Accuracy: Better than 1%

Reference: **Isida 1971a,b**

$$K_I = \sigma \sqrt{\pi a} \cdot F\left(\frac{a}{b}\right)$$

$$F\left(\frac{a}{b}\right) = 1 - .344\left(\frac{a}{b}\right)^2 - .156\left(\frac{a}{b}\right)^3 \quad \left(\frac{a}{b} \leq .8\right)$$

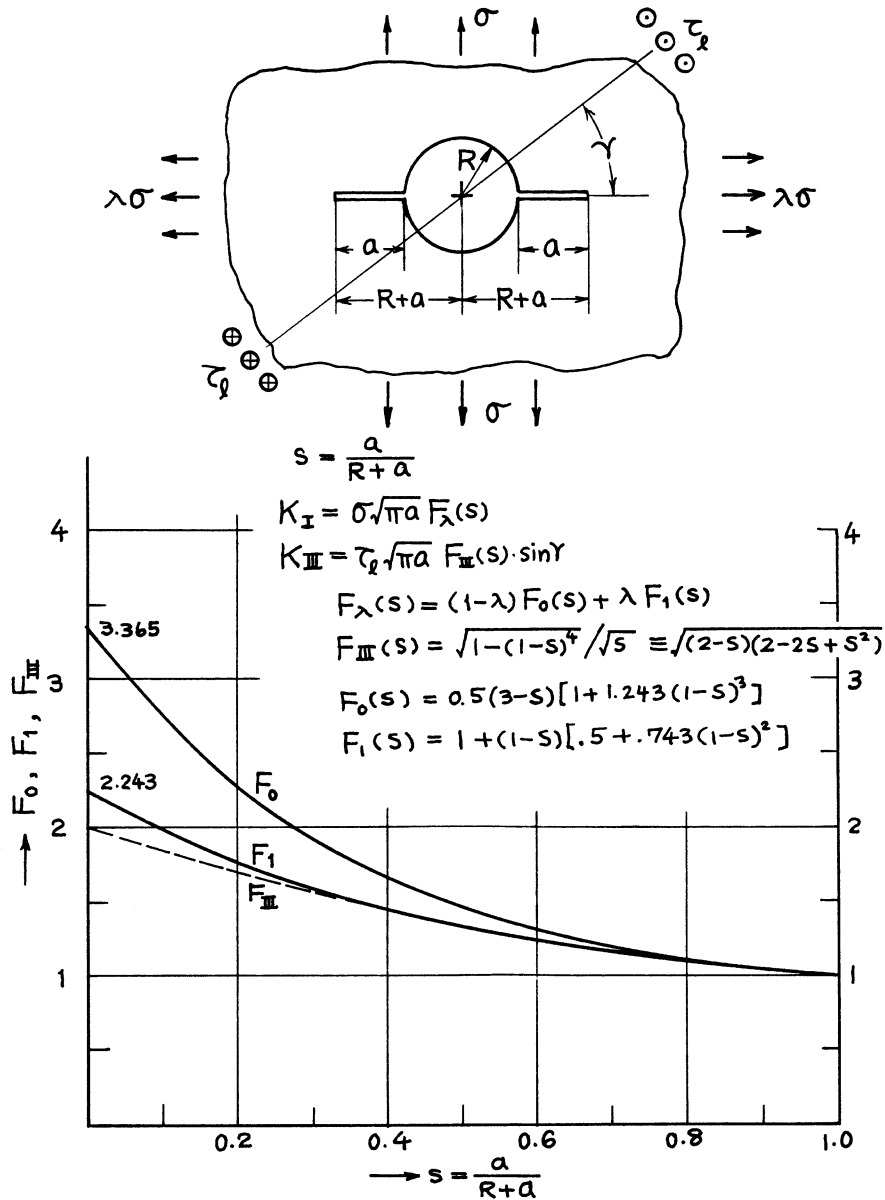


Method: Series Expansions of Complex Potentials

Accuracy: Curve is based on numerical values with 0.1% accuracy

Formula 1% for $a/b \leq 0.8$

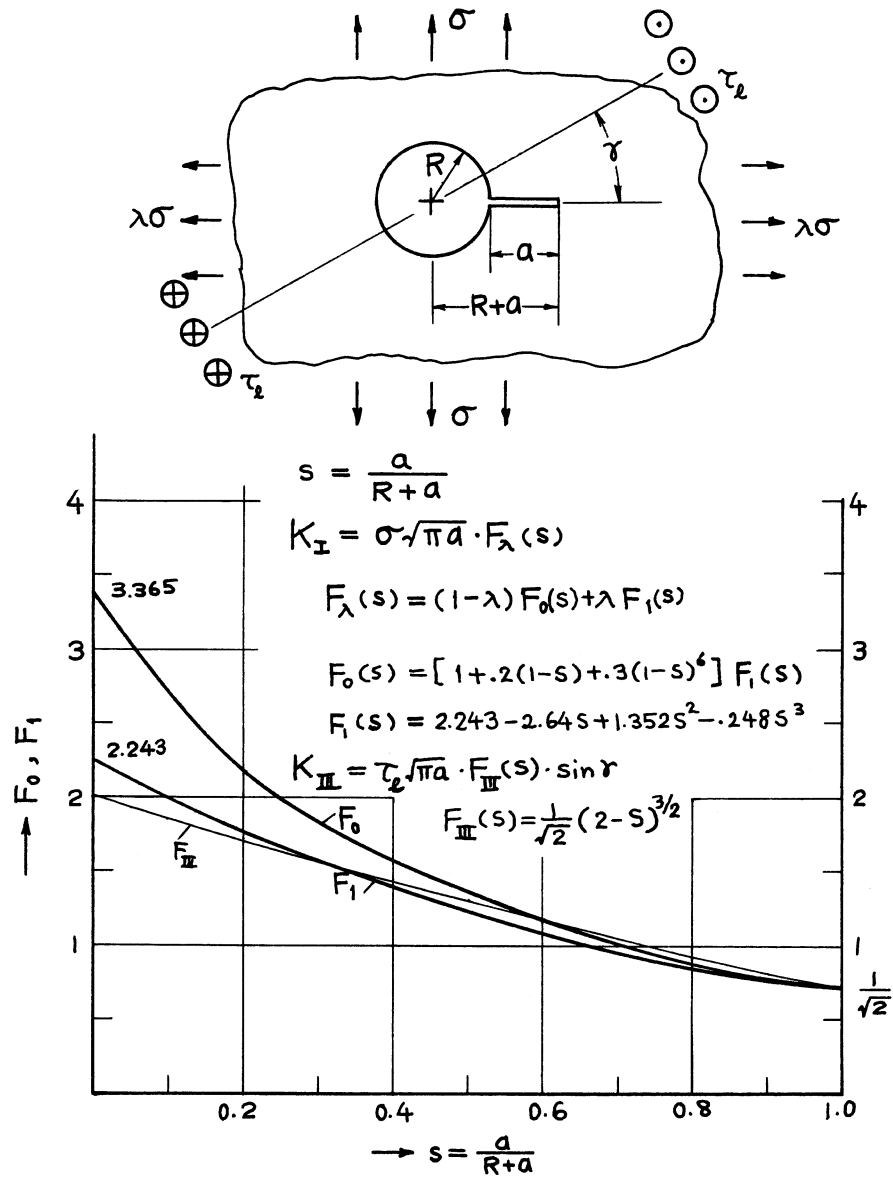
References: **Isida 1970a; Tada 1985**



Methods: Mapping Function Methods (Bowie — Mode I; Sih — Mode III), Boundary Collocation Method (Newman)

Accuracy: F_0 and F_1 curves are based on numerical values with expected accuracy of 0.1%. Formulas F_0 and F_1 1%; F_{III} Exact

References: Bowie 1956; Sih 1965a; Newman 1971; Tada 1985

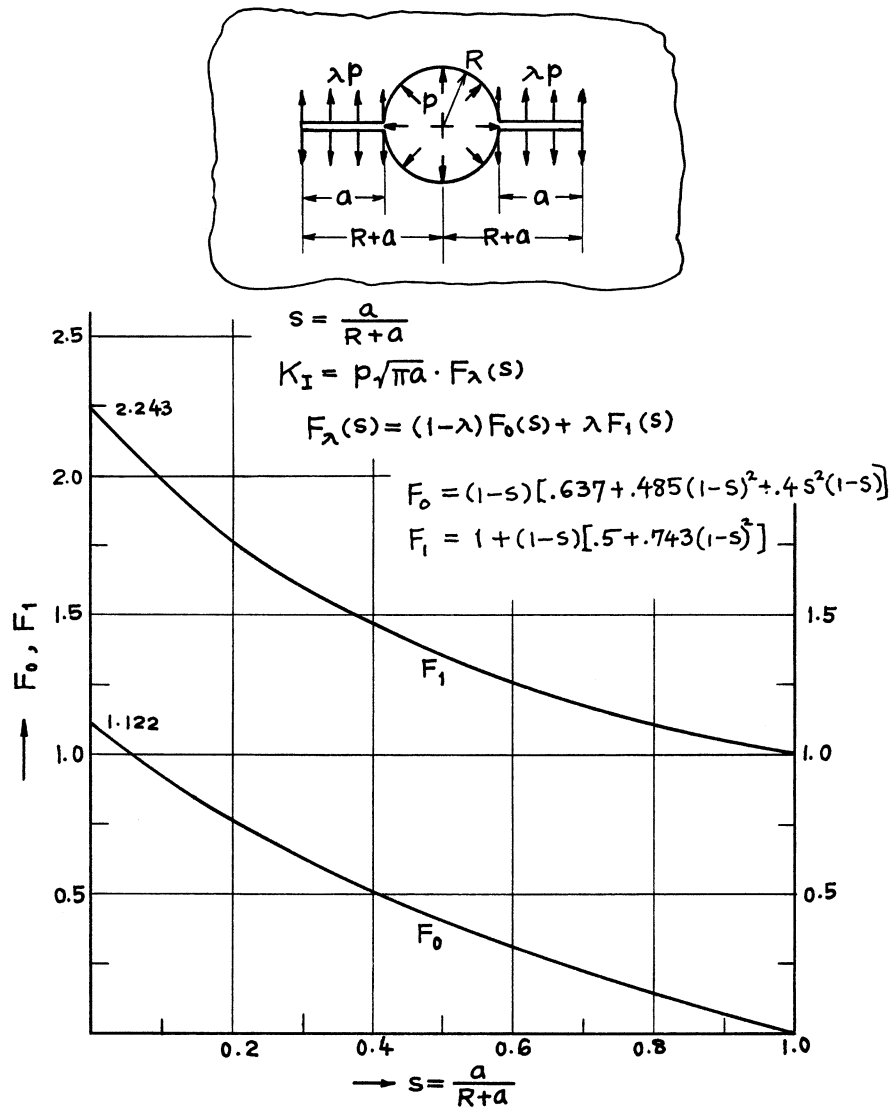


Method: Mapping Function Method

Accuracy: F_0 and F_I Better than 1%

F_{II} Exact

References: **Bowie 1956; Yokobori 1972 (or Kamei 1974); Tada 1985**

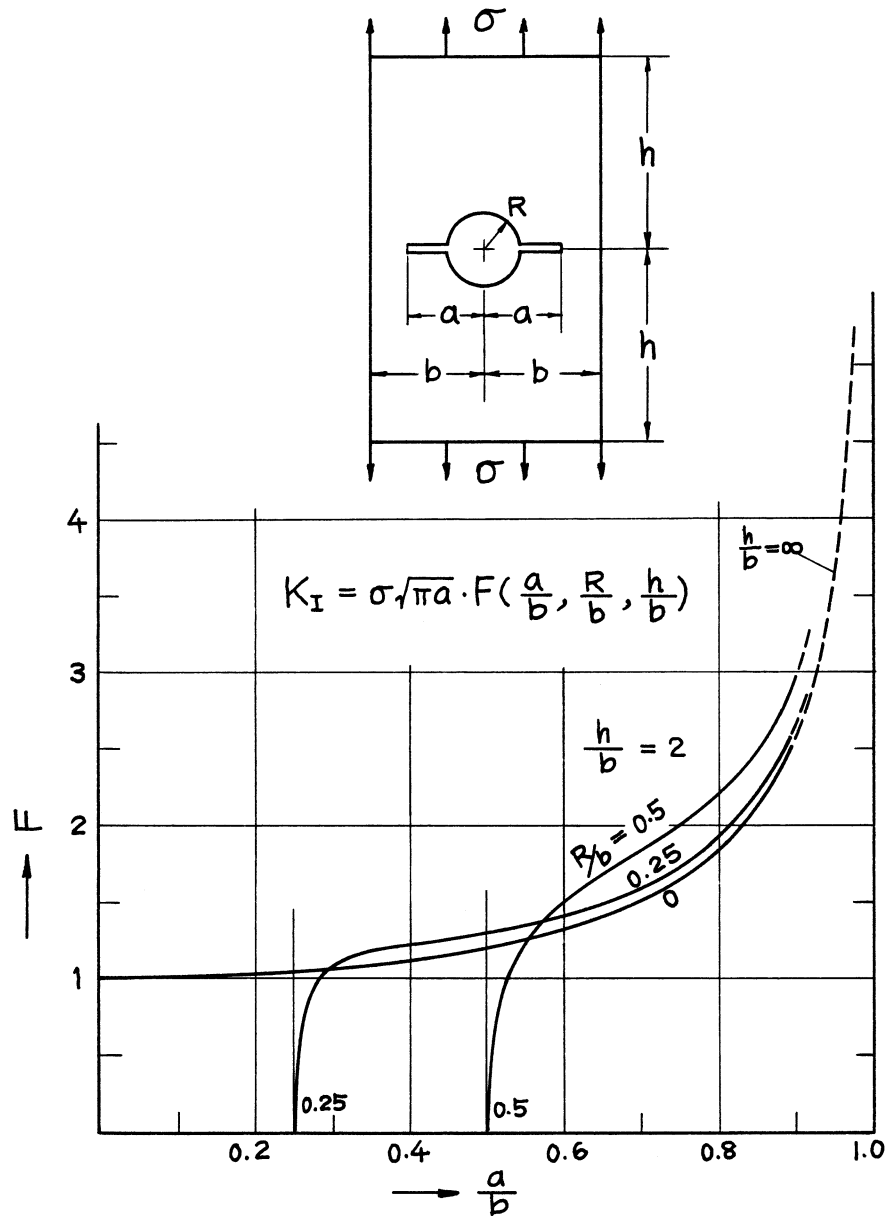


Method: Boundary Collocation Method

Accuracy: Curves are based on numerical values with 0.1% accuracy.

Formulas F_0 and F_1 1%

References: Newman 1971; Tada 1985

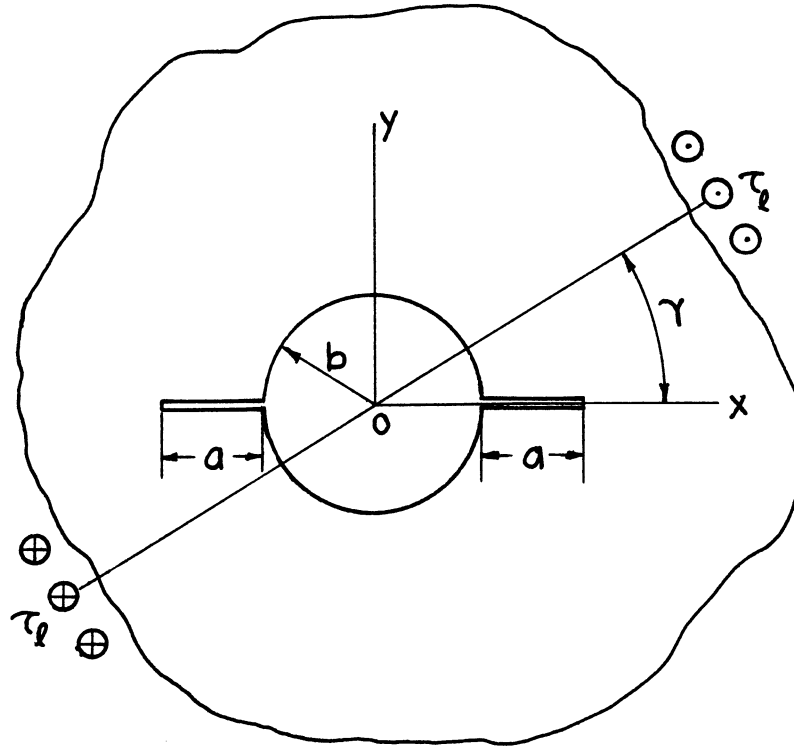


Method: Boundary Collocation Method.

Accuracy: Curves were drawn based on the results having better than 0.1% accuracy.

Reference: **Newman 1971**

See also **page 19.11**.



$$K_{III} = \tau_{\ell} \sqrt{\pi a} \frac{\sqrt{\left(1 + b/a\right)^4 - \left(b/a\right)^4}}{\left(1 + b/a\right)^{3/2}} \sin \gamma$$

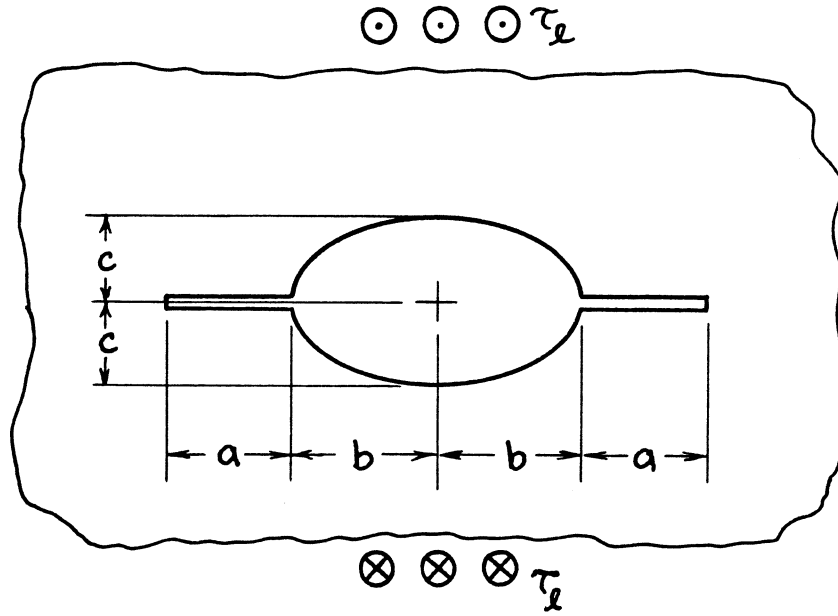
$$K_{III} \left(b/a \rightarrow \infty \right) = 2\tau_{\ell} \sqrt{\pi a} \sin \gamma$$

$$K_{III} \left(b/a \rightarrow 0 \right) = \tau_{\ell} \sqrt{\pi a} \sin \gamma$$

Method: Conformal Mapping

Accuracy: Exact

Reference: **Sih 1965a**



$$K_{III} = \tau_\ell \sqrt{\pi(b+c)} \cdot \sqrt{\frac{s^4 - 1}{2s \left(s^2 - \frac{b-c}{b+c} \right)}}$$

where

$$s = \frac{b+a + \sqrt{(b+a)^2 - (b^2 - c^2)}}{b+c}$$

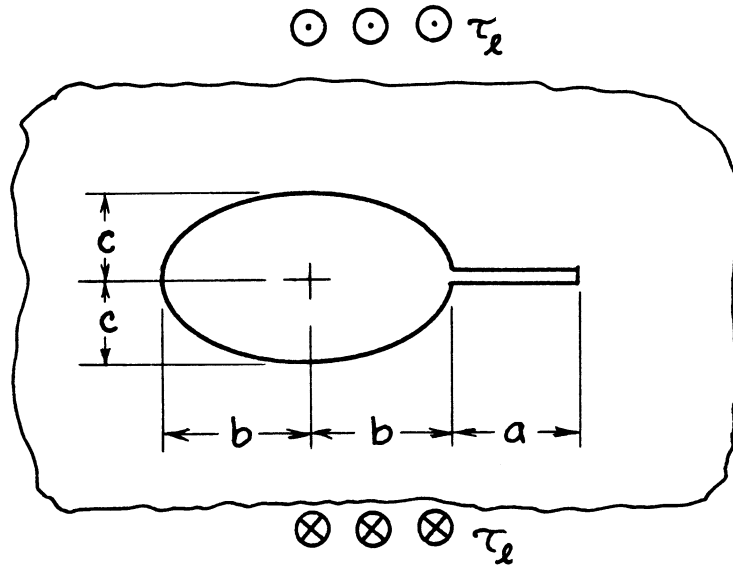
For Circular Hole,

$$c = b = R : K_{III} = \tau_\ell \sqrt{\pi(R+a)} \sqrt{1 - \left(\frac{R}{R+a} \right)^4}$$

Method: Conformal Mapping

Accuracy: Exact

Reference: **Yokobori 1972 (or Kamei 1974)**



$$K_{III} = \tau_\ell \sqrt{\pi(b+c)} \frac{s+1}{2} \sqrt{\frac{s^2-1}{s\left(s^2 - \frac{b-c}{b+c}\right)}}$$

where

$$s = \frac{b+a + \sqrt{(b+a)^2 - (b^2 - c^2)}}{b+c}$$

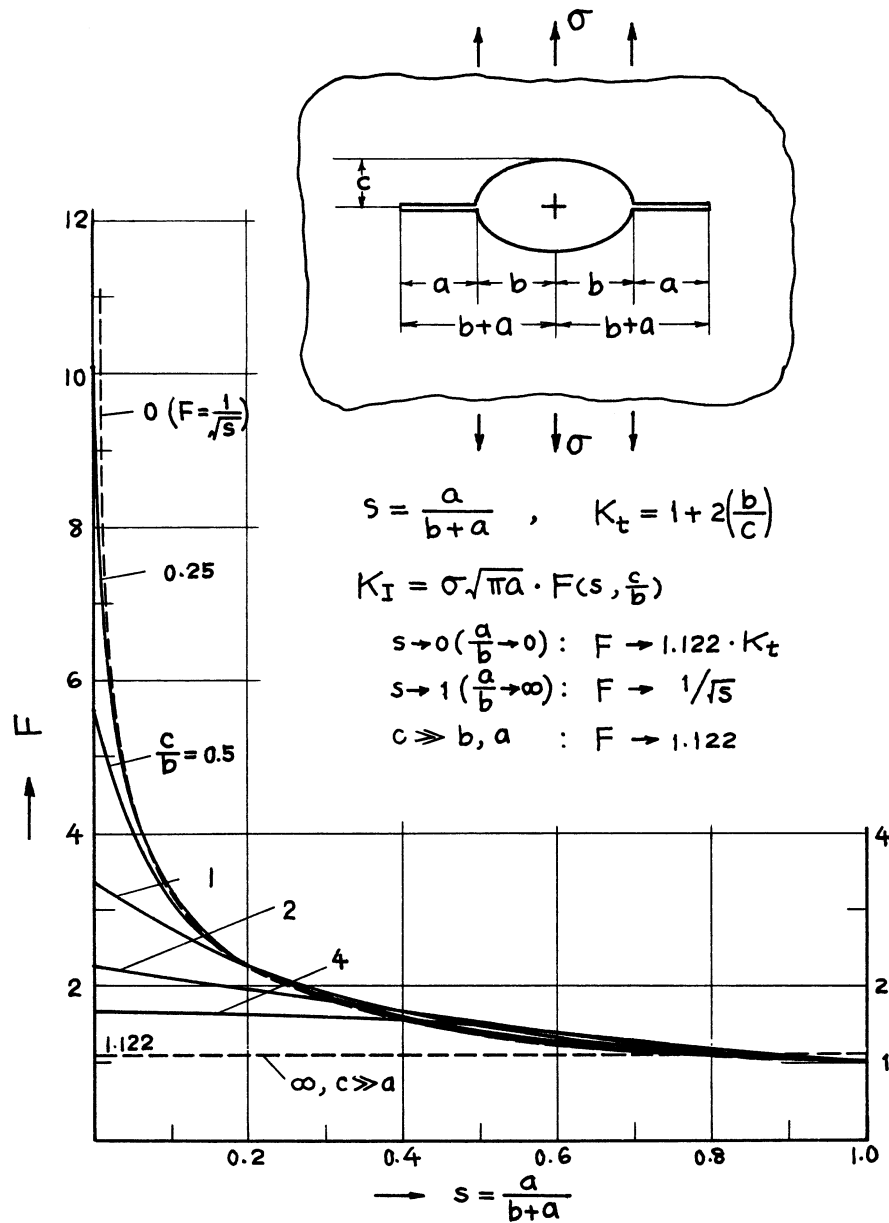
For Circular Hole,

$$c = b = R : K_{III} = \tau_\ell \sqrt{\pi a} \cdot \frac{1}{\sqrt{2}} \left(1 + \frac{R}{R+a}\right)^{3/2}$$

Method: Conformal Mapping

Accuracy: Exact

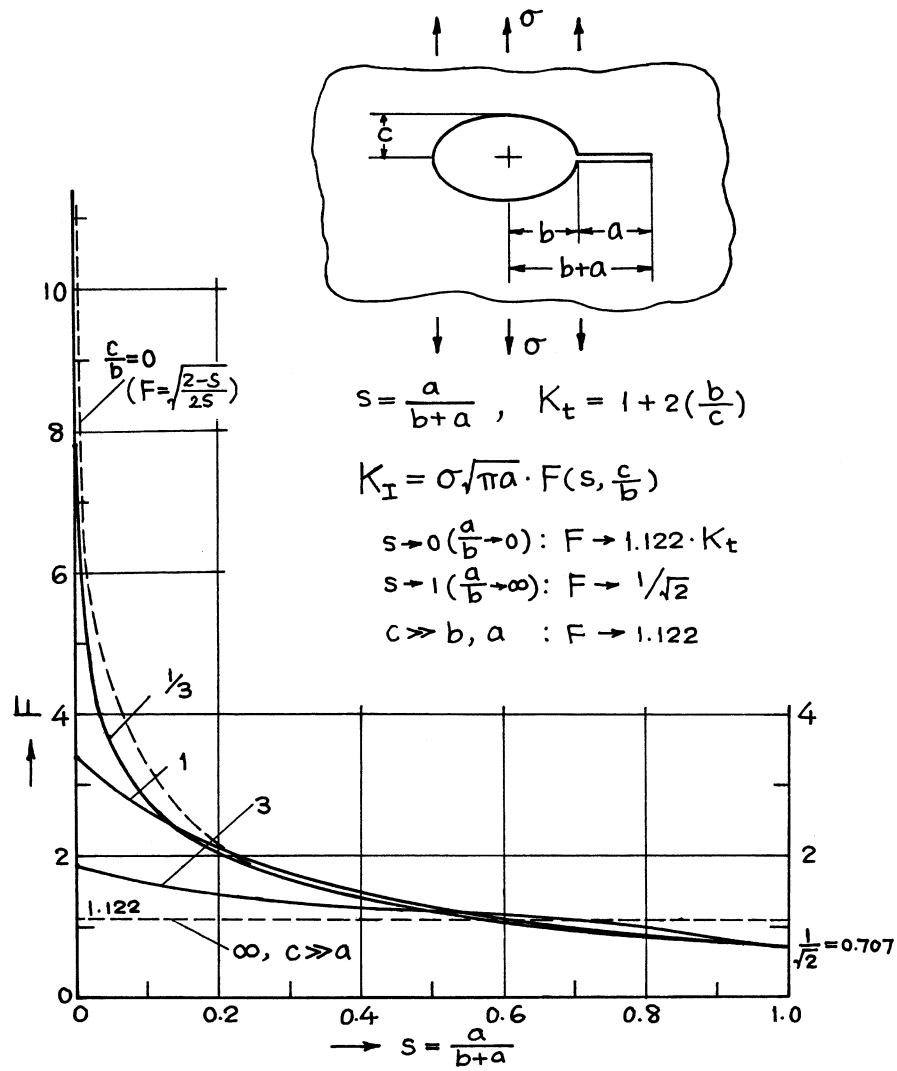
Reference: Yokobori 1972 (or Kamei 1974)



Methods: Boundary Collocation Method (Newman), Body Force Method (Nishitani)

Accuracy: Curves were drawn based on the results having better than 0.1% accuracy

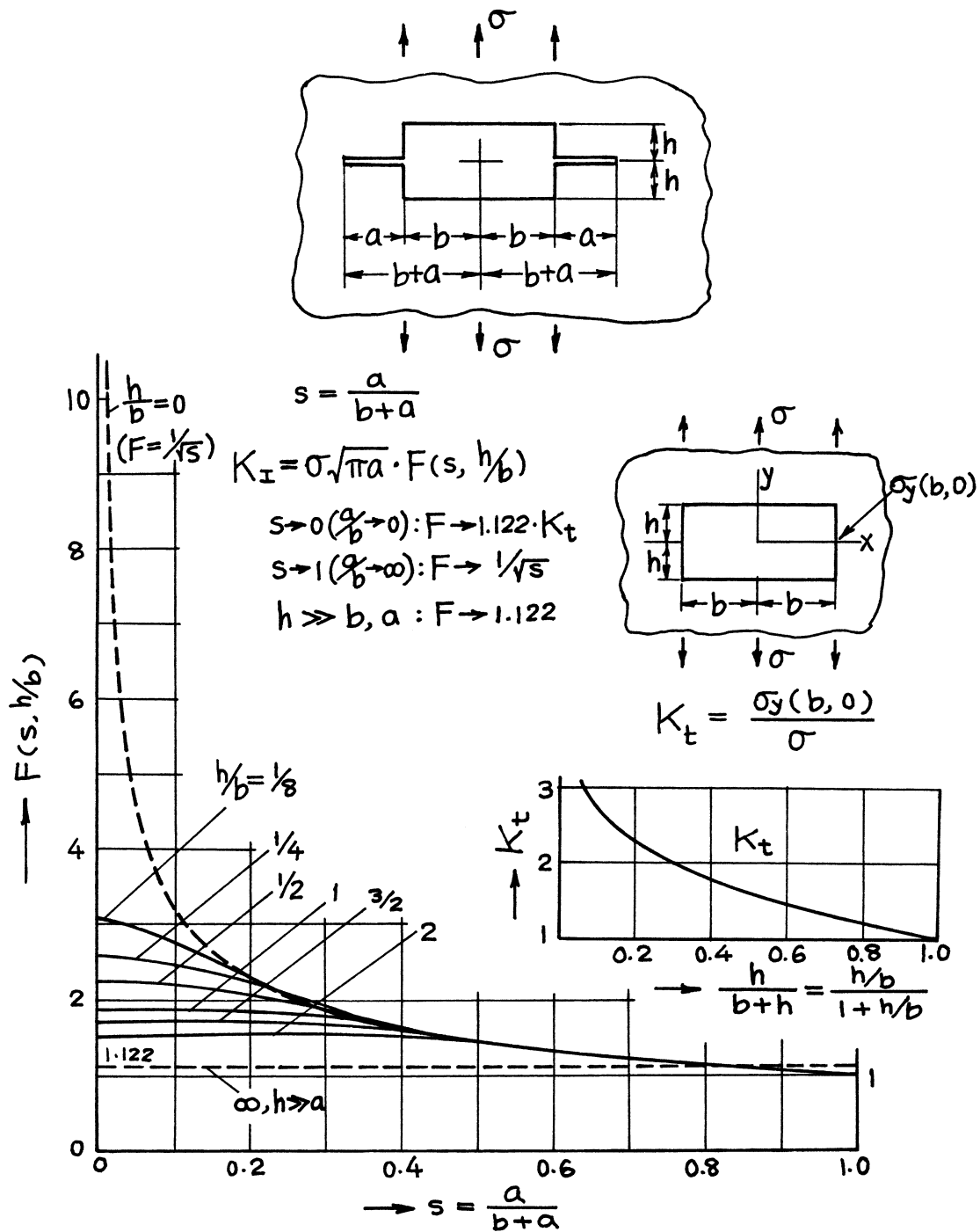
References: Nishitani 1969; Newman 1971



Method: Integral Equation

Accuracy: 3%

Reference: **Berezhnitskii 1966**



Methods: Mapping Collocation Method (Neal, $s \geq 0.25$), Estimated (Tada, $s < 0.25$)

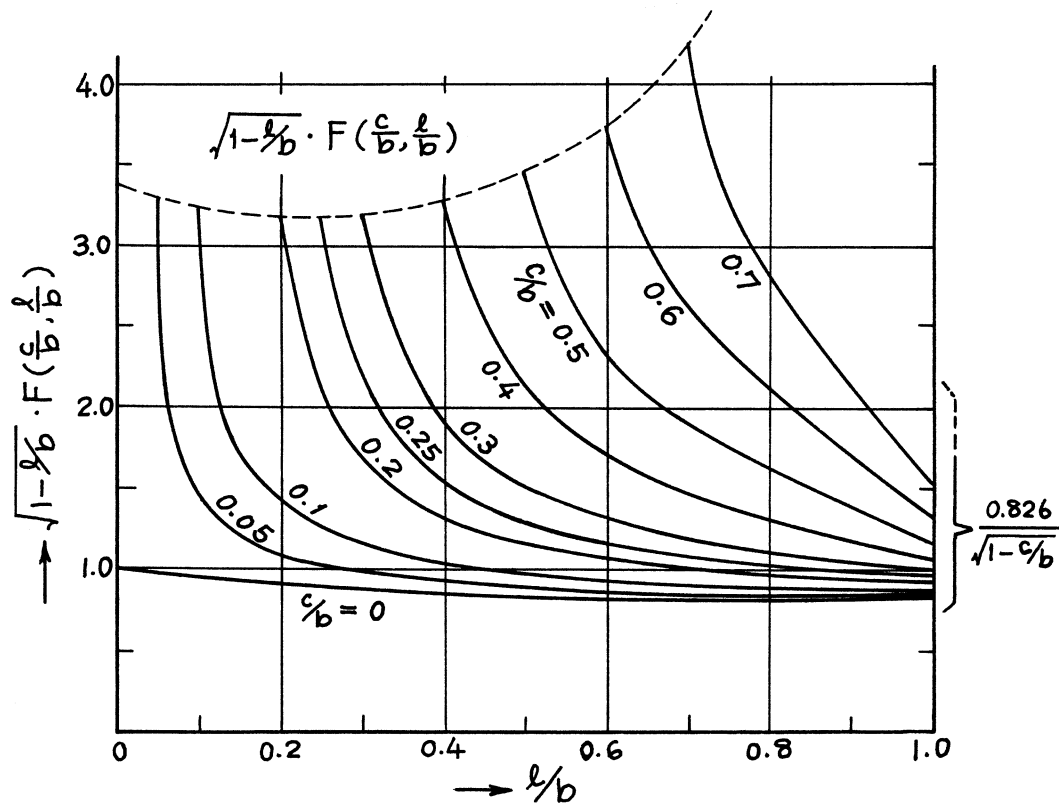
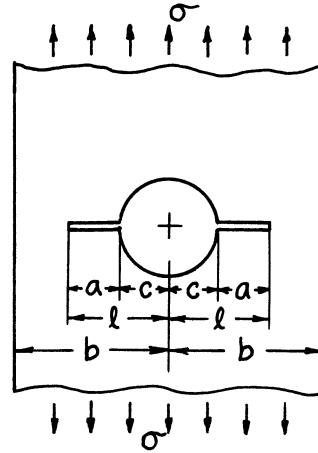
Accuracy: Expected to be better than 5% for $s < 0.25$, better than 2% for $s \geq 0.25$

References: Neal 1970; Tada 1973; Savin 1961, 1968 (K_t values)

NOTE: Neal's results for small cracks ($s < 0.25$) appear to be too large.

$$\ell = c + a$$

$$K_I = \sigma \sqrt{\pi a} F\left(\frac{c}{b}, \frac{\ell}{b}\right)$$



Methods: Boundary Collocation ($c/b = 0.25, 0.5$; Newman), Estimated by Interpolation (c/b other than 0.25, 0.5; Tada)

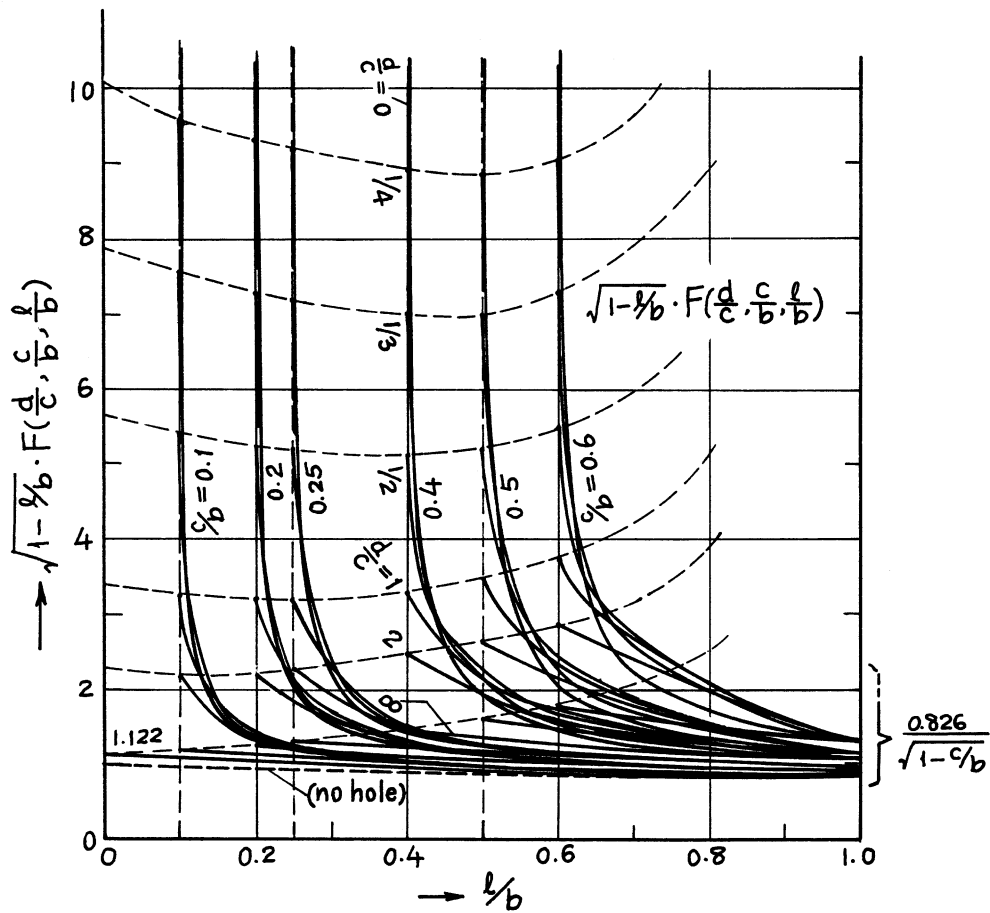
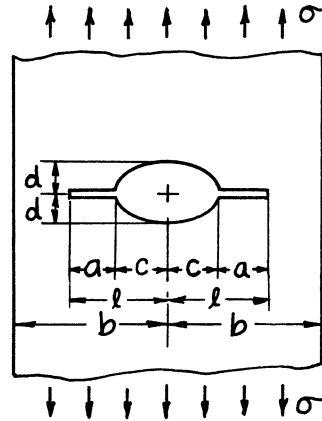
Accuracy: Accurate for $c/b = 0.25, 0.5$; better than 5% for other values of c/b

References: **Newman 1971; Tada 1973**

NOTE: For $c/b \ll 1$, see **page 19.1 and 2.1**. See also **p. 19.4**.

$$\ell = c + a$$

$$K_I = \sigma \sqrt{\pi a} \cdot F\left(\frac{d}{c}, \frac{c}{b}, \frac{\ell}{b}\right)$$

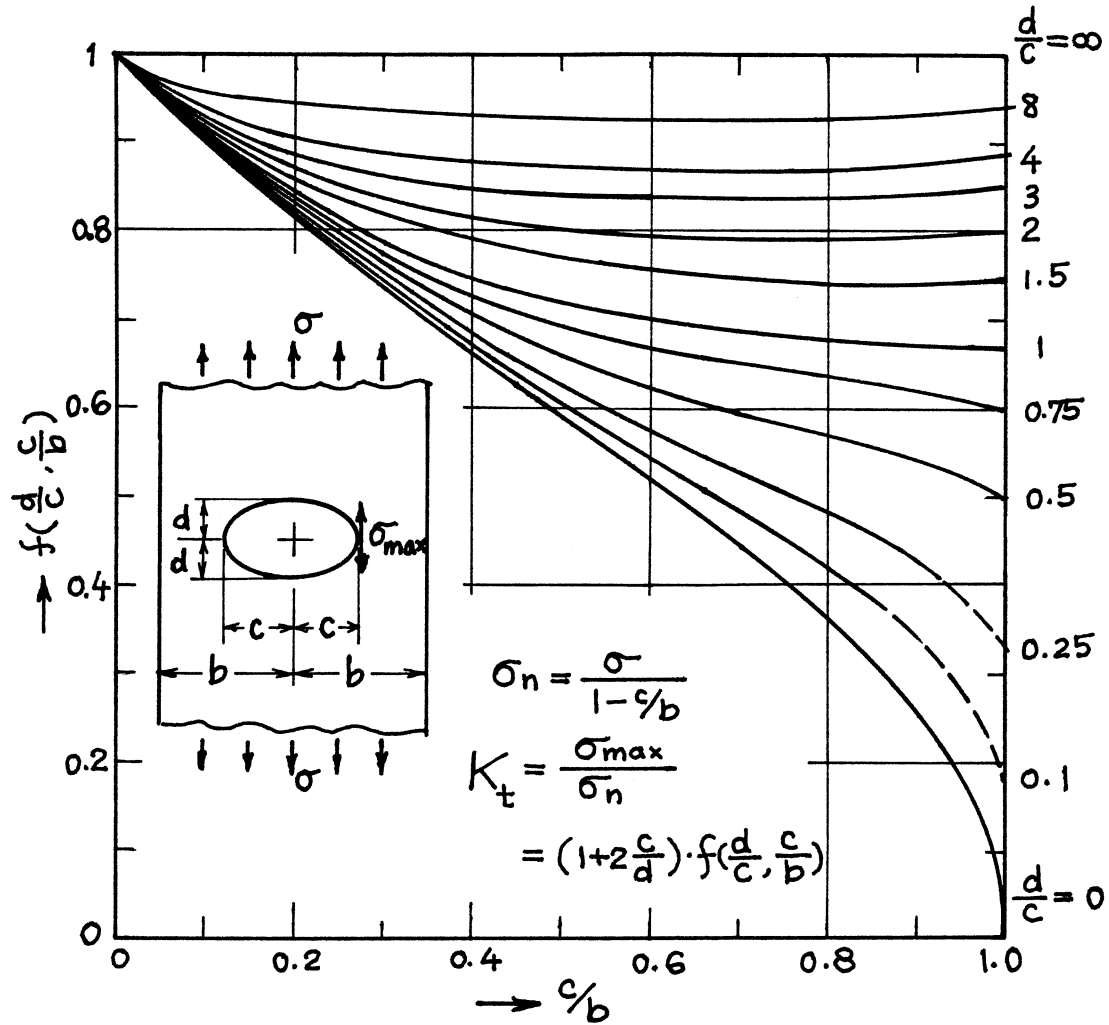


Method: Estimated by Interpolation

Accuracy: Better than 5 % for any d/c , c/b , and ℓ/b

Reference: **Tada 1973**

NOTE: For $c/b \ll 1$, see **page 19.8**.



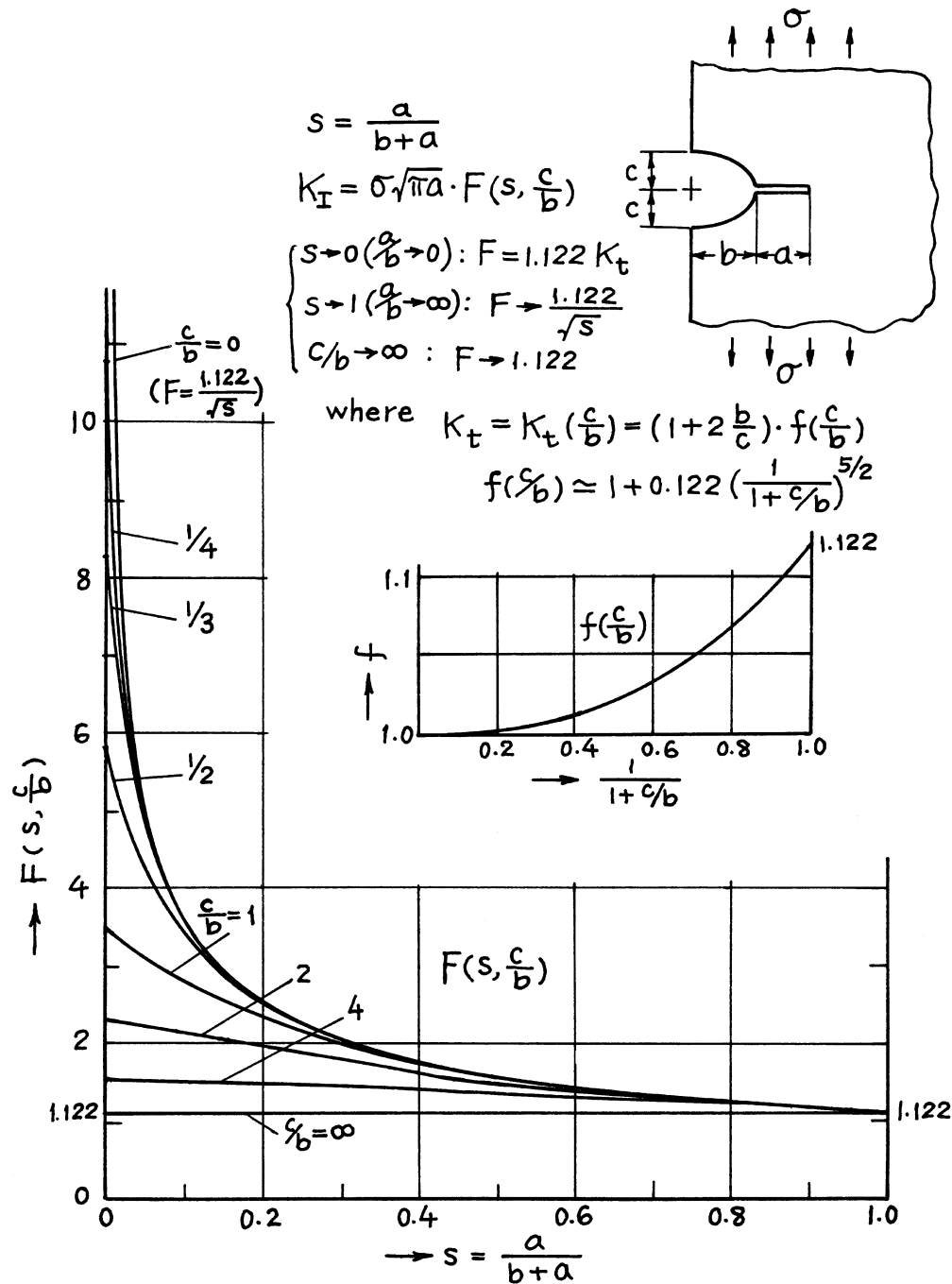
Method: Estimated by Interpolation

Accuracy: $f(d/c = \infty, c/b)$ Exact

$f(d/c = 1, c/b), f(d/c \rightarrow 0, c/b)$ Accurate

For all other values of d/c , accuracy is expected to be much better than 5 %.

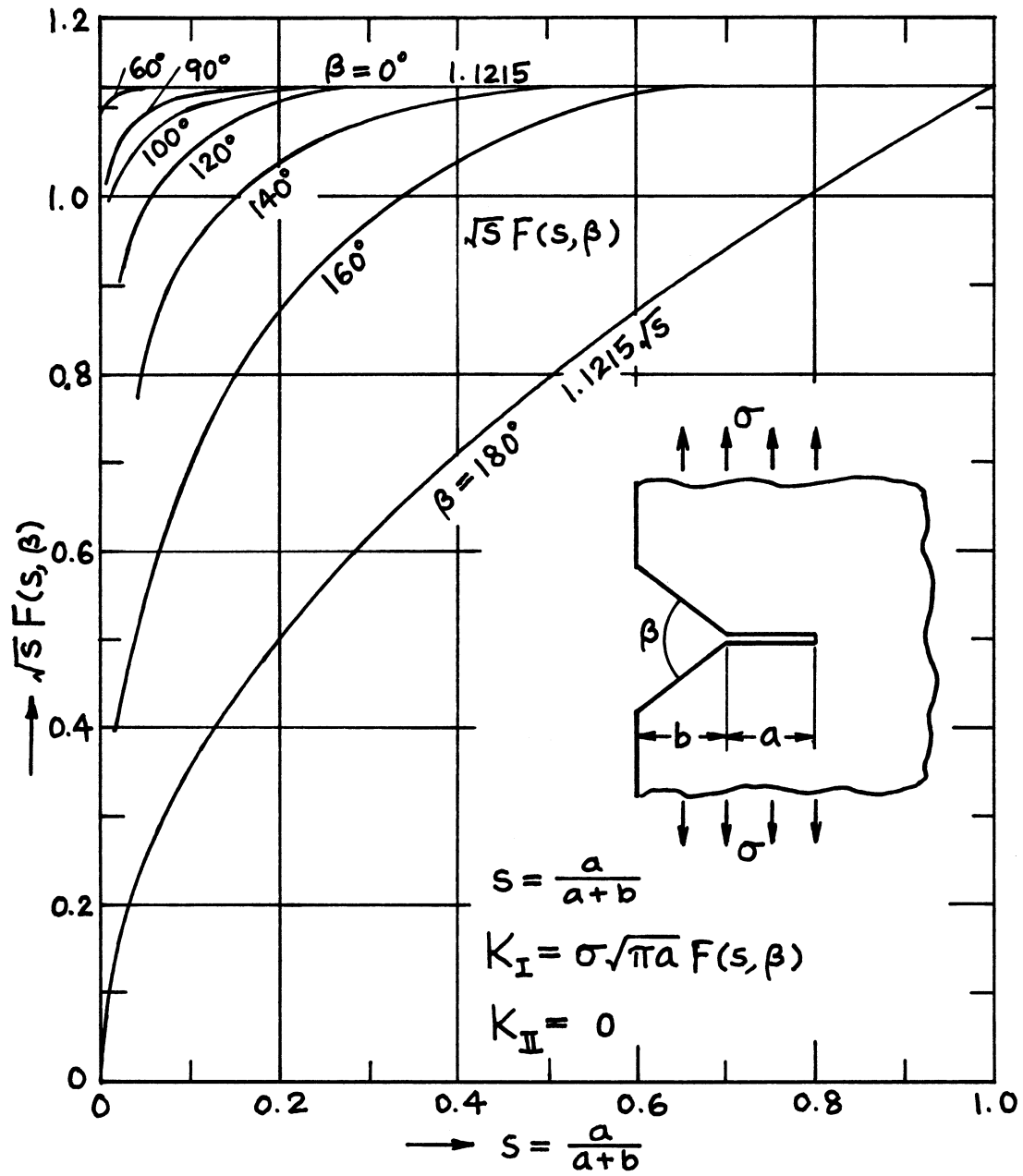
References: **Tada 1974**



Methods: Stress Relaxation (Superposition) (Nishitani; $c/b = 1/2, 1, 2$ and $0.2 \leq a/b \leq 1$), Estimated by Interpolation (Tada)

Accuracy: Better than 2%

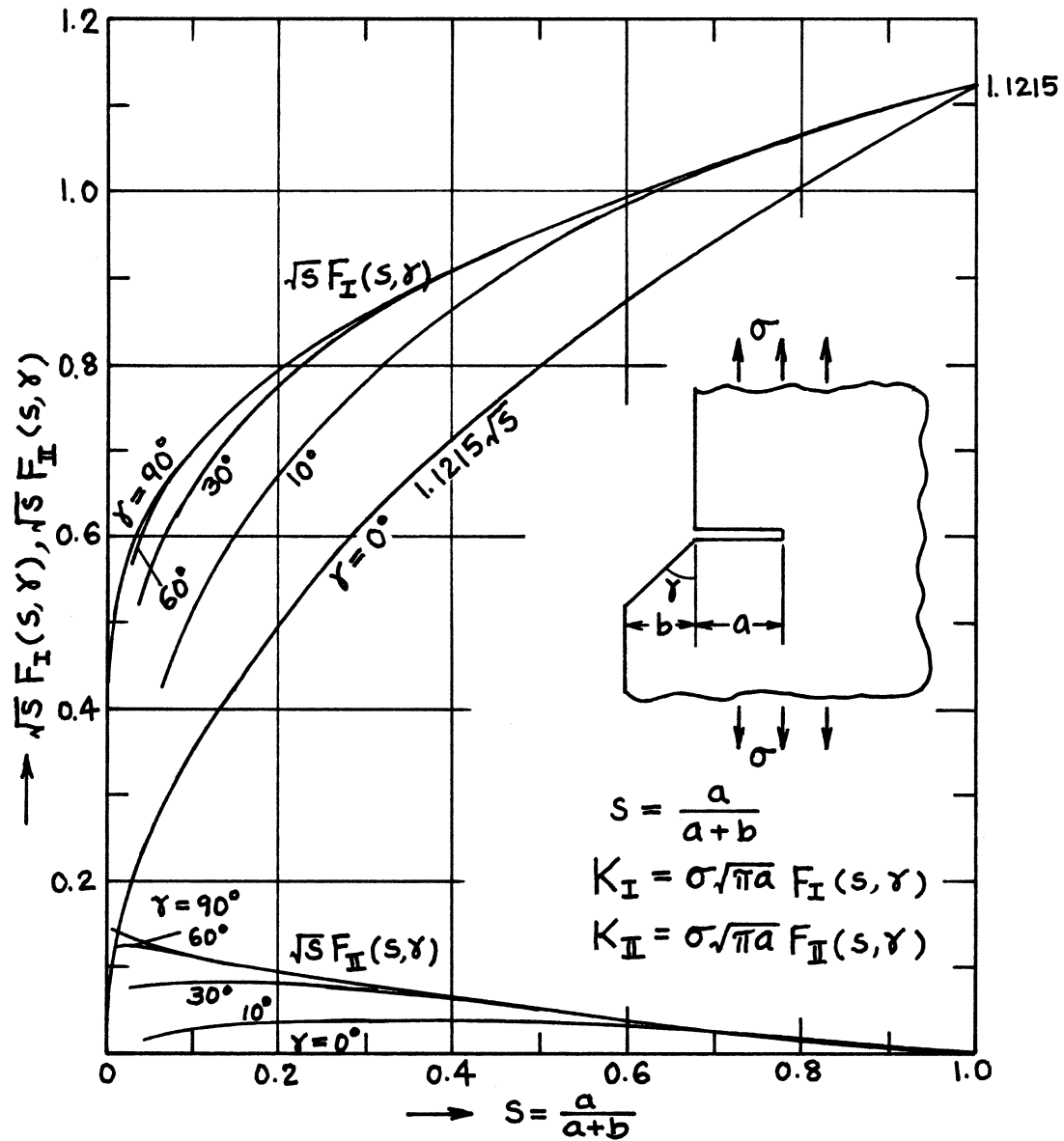
References: Nishitani 1973; Tada 1973



Method: Conformal Mapping

Accuracy: Curves are based on accurate (0.1%) numerical values (Hasebe 1978).

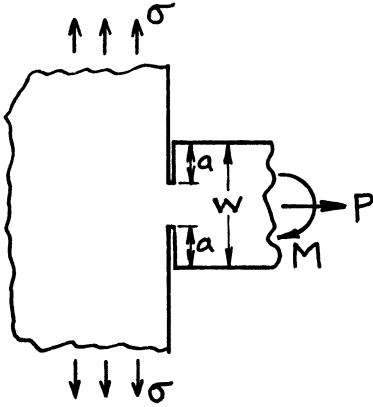
References: Hasebe 1978; Tada 2000



Method: Conformal Mapping

Accuracy: Curves are based on accurate (0.1%) numerical values (Hasebe 1980).

References: Hasebe 1980; Tada 2000



$$A = 2a/W$$

$$K_I = \sqrt{\pi a} \left[\sigma F_{I\sigma}(A) + \frac{P}{W} F_{IP}(A) + \frac{M}{W^2} F_{IM}(A) \right]$$

$$K_{II} = \sqrt{\pi a} \left[\sigma F_{II\sigma}(A) + \frac{P}{W} F_{IIP}(A) + \frac{M}{W^2} F_{IIM}(A) \right]$$

$$F_{I\sigma}(A) = \sqrt{\frac{1-A}{A}} \cdot 0.15 \left[(1-A)^5 - A(1-A)^8 \right]$$

$$F_{II\sigma}(A) = \sqrt{\frac{1-A}{A}} \cdot \left[0.11 + 0.14 \sqrt{1 - (1-A)^2} - 0.016(1-A)^8 \right]$$

$$F_{IP}(A) = \frac{1}{\sqrt{A(1-A)}} \cdot \left[0.637 - 0.224(1-A)^4 + 0.75\sqrt{A}(1-A)^3(0.3-A) \right]$$

$$F_{IIP}(A) = \frac{1}{\sqrt{A(1-A)}} \cdot 0.145(1-A)$$

$$F_{IM}(A) = \frac{1}{\sqrt{A}(1-A)^{3/2}} \cdot \left[0.424 - 0.15(1-A)^4 + 0.2\sqrt{A}(1-A)^{9/2} \right]$$

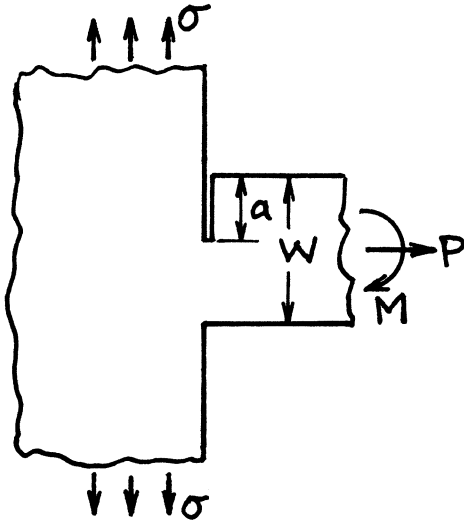
$$F_{IIM}(A) = \frac{1}{\sqrt{A}(1-A)^{3/2}} \cdot \left[0.85(1-A)^4 + 0.12A^2(1-A)^3 \right]$$

Method: Conformal Mapping

Accuracy: F_{IP} , F_{IM} 0.5%; $F_{II\sigma}$, F_{IIM} 1%; $F_{I\sigma}$, F_{IIP} 2%

References: **Hasebe 1981; Tada 2000**

NOTE: All approximate formulas (**Tada 2000**) are based on accurate (0.1%) numerical values (**Hasebe 1981**).



$$A = a/W$$

$$K_I = \sqrt{\pi a} \left[\sigma F_{I\sigma}(A) + \frac{P}{W} F_{IP}(A) + \frac{M}{W^2} F_{IM}(A) \right]$$

$$K_{II} = \sqrt{\pi a} \left[\sigma F_{II\sigma}(A) + \frac{P}{W} F_{IIP}(A) + \frac{M}{W^2} F_{IIM}(A) \right]$$

$$F_{I\sigma}(A) = \sqrt{\frac{1-A}{A}} \cdot \left[0.018 + 0.069 e^{-12.5\left(\frac{A}{1-A}\right)} \right]$$

$$F_{II\sigma}(A) = \sqrt{\frac{1-A}{A}} \cdot \left[0.156 - 0.067 e^{-8.9\left(\frac{A}{1-A}\right)} \right]$$

$$F_{IP}(A) = \frac{1}{\sqrt{A}(1-A)^{3/2}} \cdot \left[0.379 + 0.624A - 0.062 e^{-12\left(\frac{A}{1-A}\right)} \right]$$

$$F_{IIP}(A) = \frac{1}{\sqrt{A}(1-A)^{3/2}} \cdot \left[0.126 - 0.24A - 0.023(1-A)^5 \right]$$

$$F_{IM}(A) = \frac{1}{\sqrt{A}(1-A)^{3/2}} \cdot \left[2.005 - 0.72 e^{-9\left(\frac{A}{1-A}\right)} \right]$$

$$F_{IIM}(A) = \frac{1}{\sqrt{A}(1-A)^{3/2}} \cdot \left[-0.228 + (1-A)^4 (0.577 - 0.2A + 0.8A^2) \right]$$

Method: Conformal Mapping

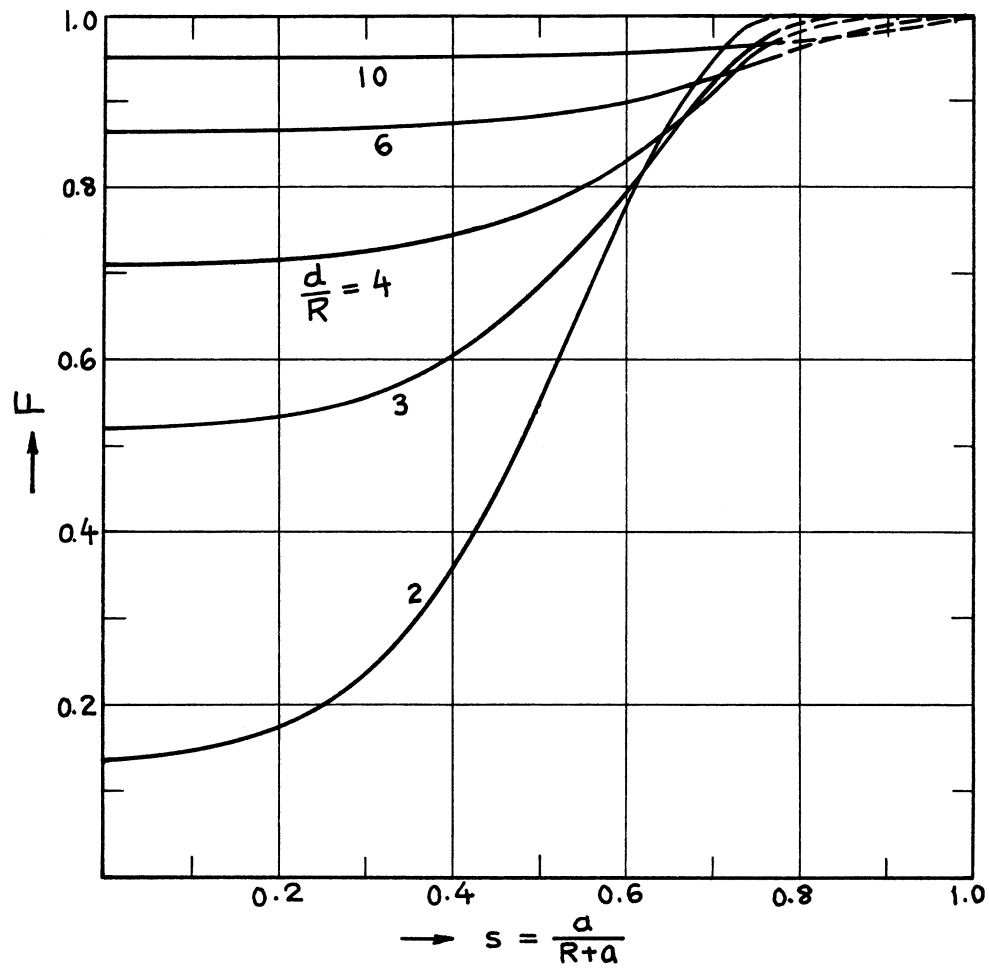
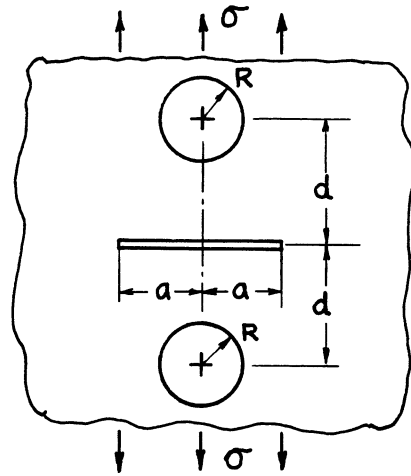
Accuracy: F_{IP} , F_{IIP} , F_{IIM} 1%; $F_{I\sigma}$, $F_{II\sigma}$, F_{IM} 2%

References: **Hasebe 1987; Tada 2000**

NOTE: All approximate formulas (**Tada 2000**) are based on accurate (0.1%) numerical values (**Hasebe 1987**).

$$s = \frac{a}{R+a}$$

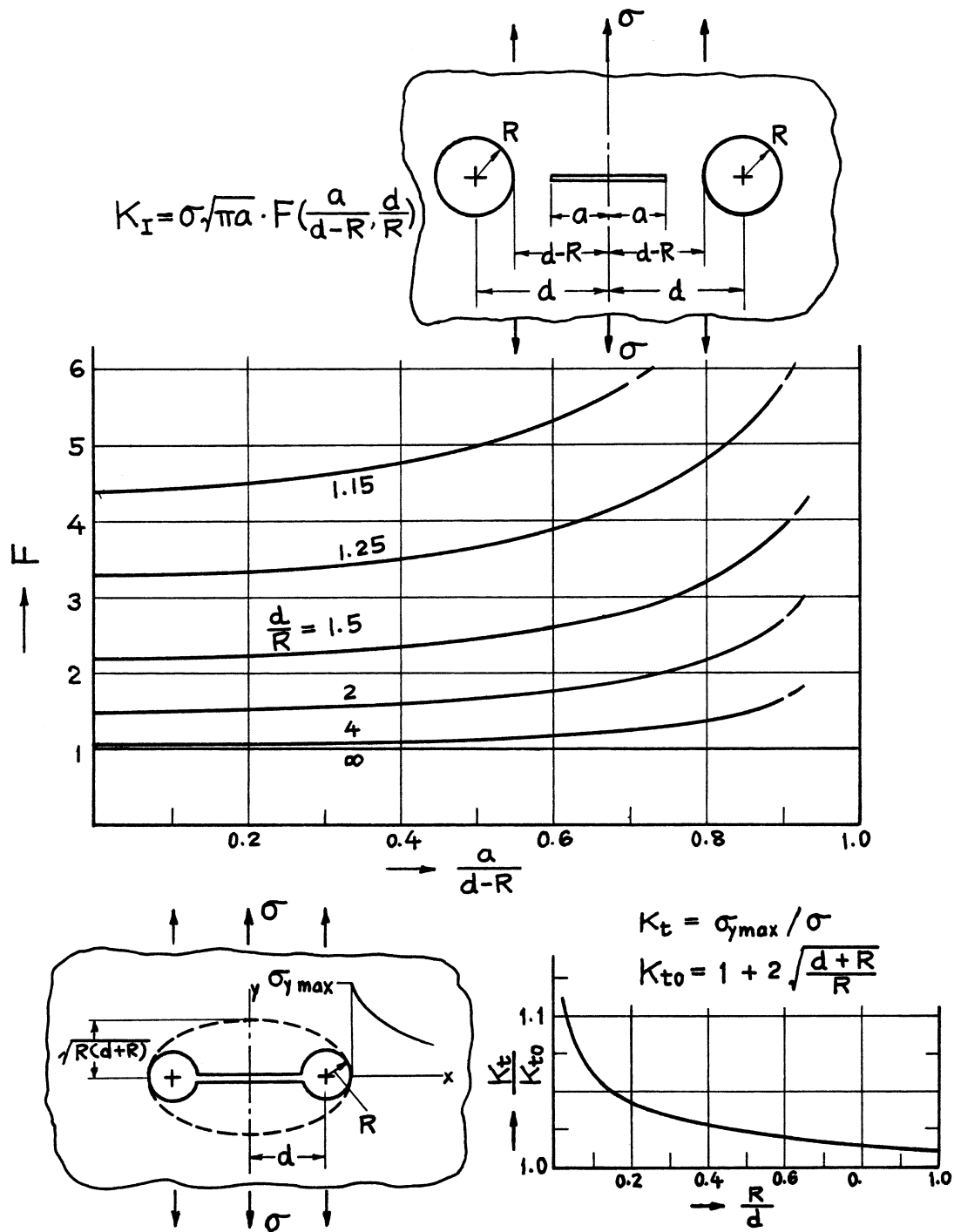
$$K_I = \sigma\sqrt{\pi a} \cdot F\left(s, \frac{d}{R}\right)$$



Method: Boundary Collocation Method

Accuracy: Curves (solid lines) were drawn based on the results having 0.1% accuracy.

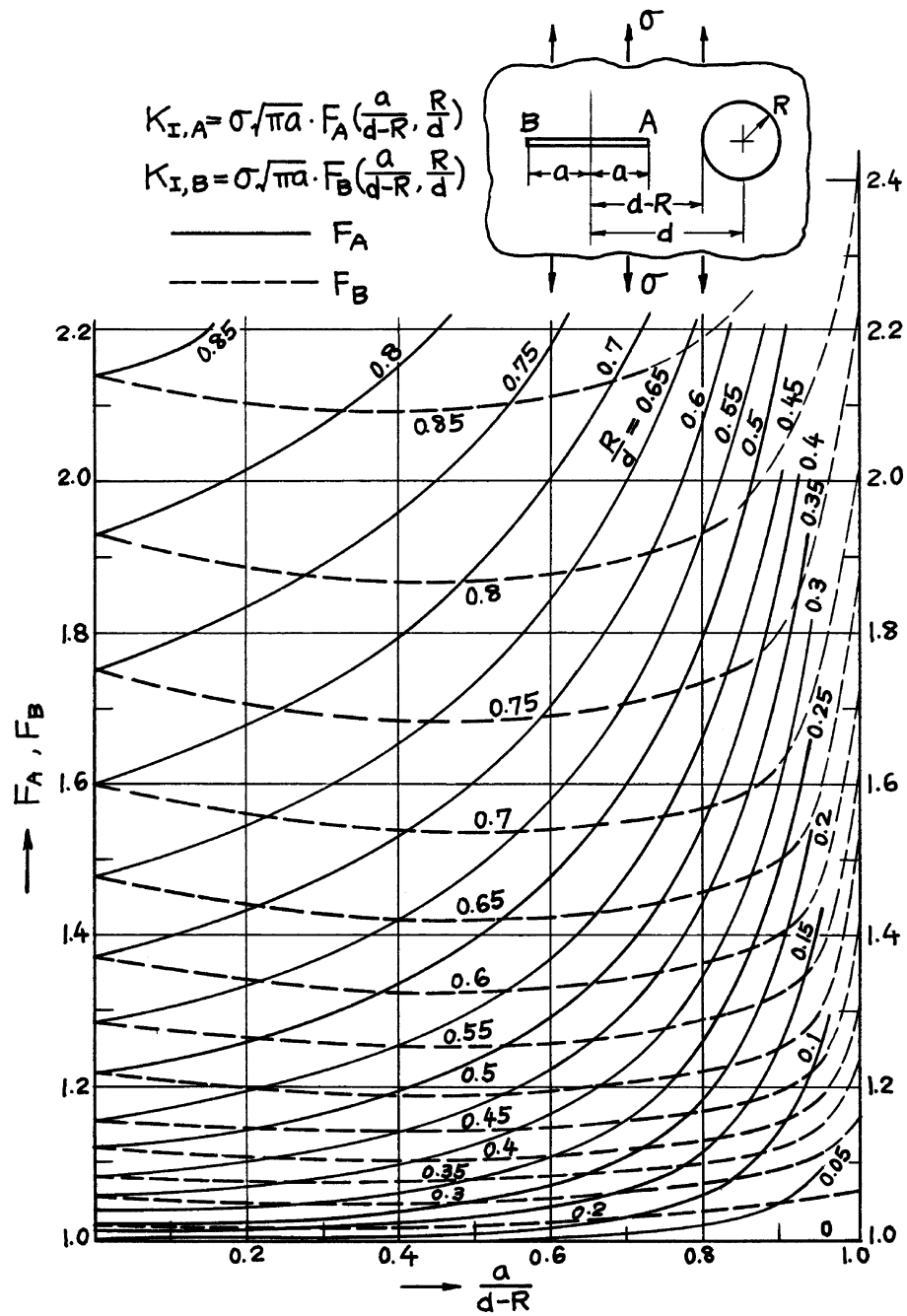
Reference: **Newman 1971**



Methods: Boundary Collocation Method (Newman), Expansions of Complex Stress Potentials (Isida)

Accuracy: Curves were drawn based on the results having 0.1% accuracy.

References: **Newman 1971; Isida 1973**

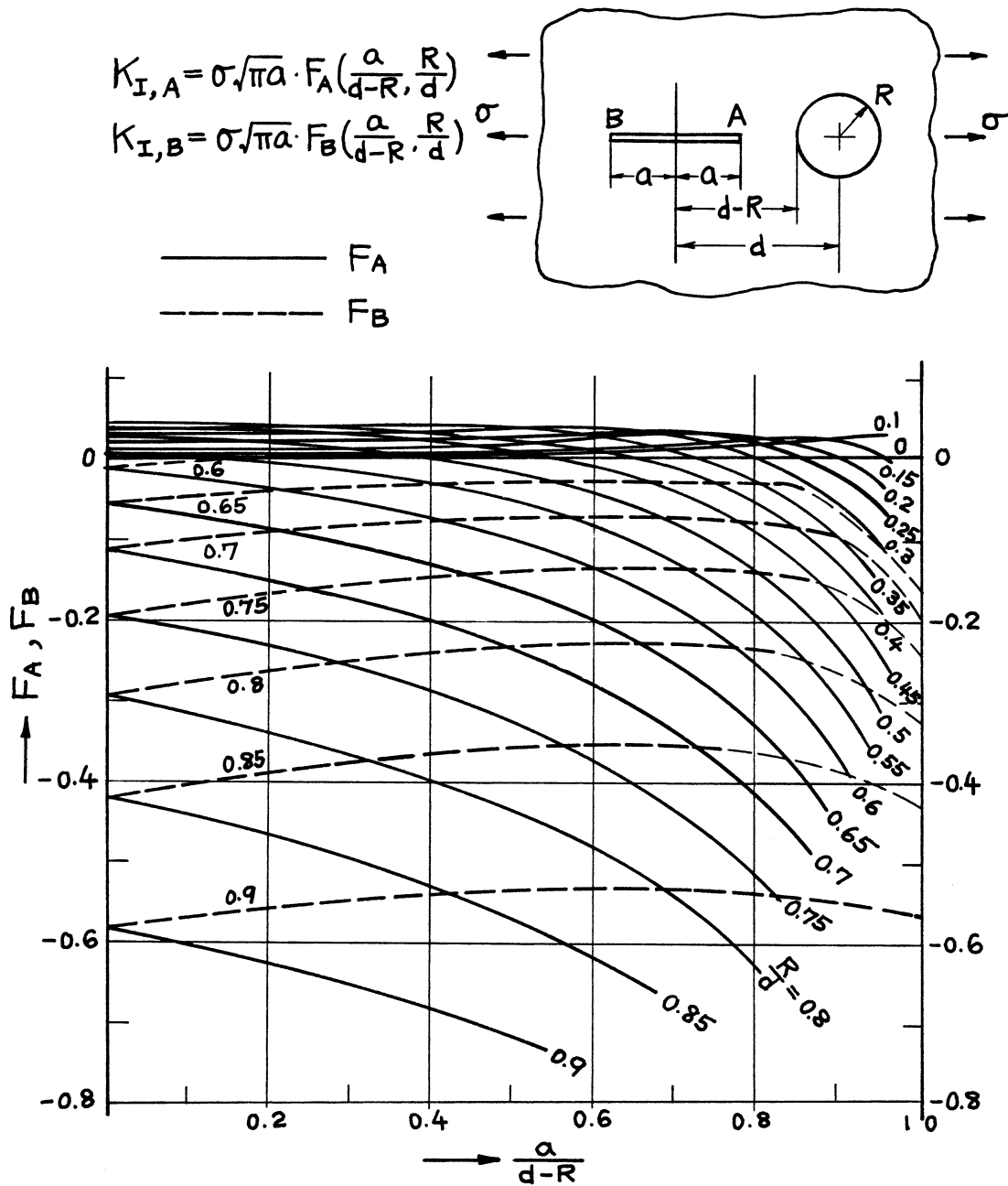


Method: Expansions of Complex Stress Potentials

Accuracy: Curves were drawn based on the results having 0.1% accuracy (thick solid and dashed lines).

Reference: **Isida 1970a**

NOTE: The values of F_B for $a/(d-R) \rightarrow 1$ were taken from p. 19.2 where $\lambda = 0$.

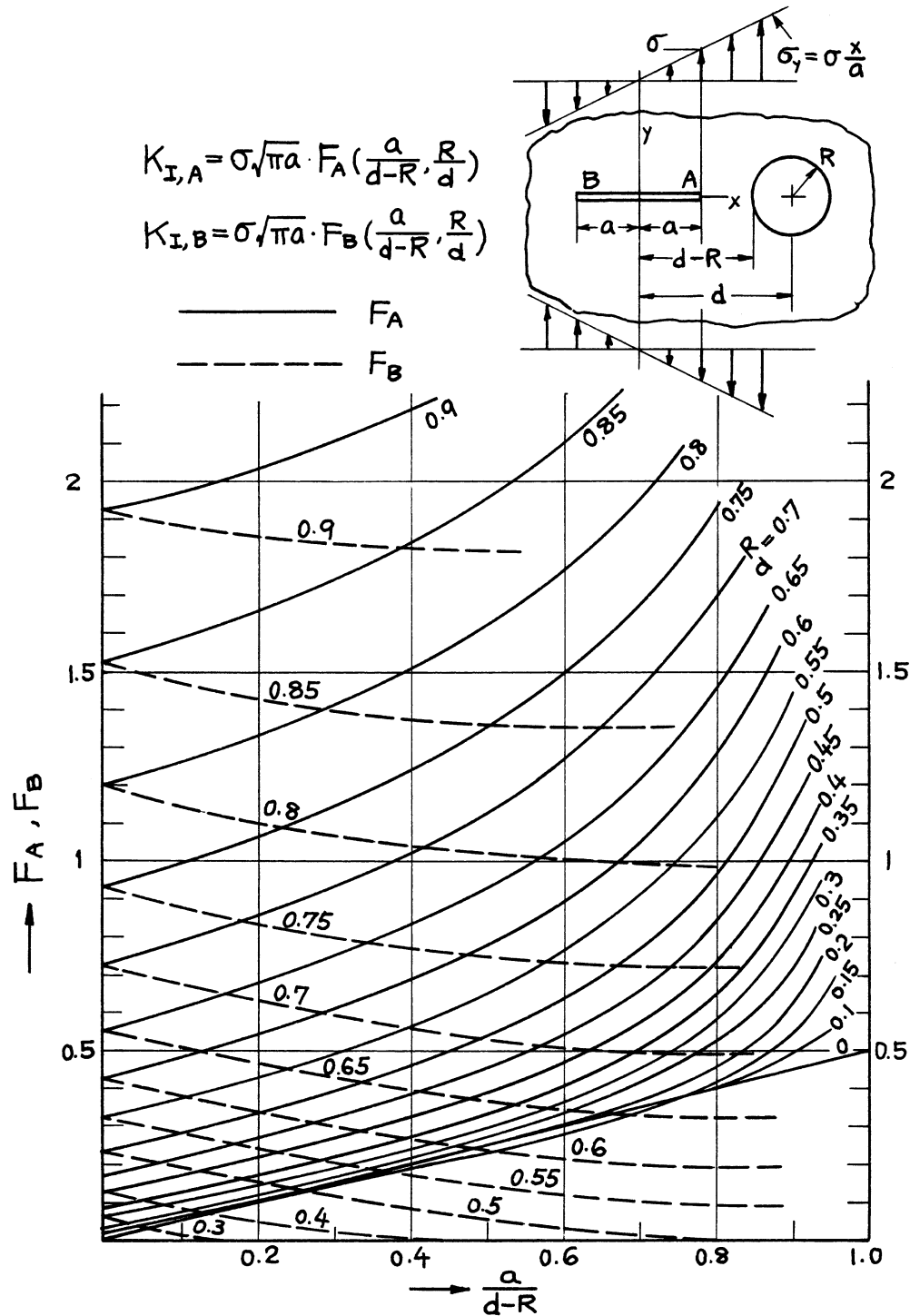


Method: Expansions of Complex Stress Potentials

Accuracy: Curves were drawn based on the results having 0.1% accuracy (thick solid and dashed lines).

Reference: **Isida 1970a**

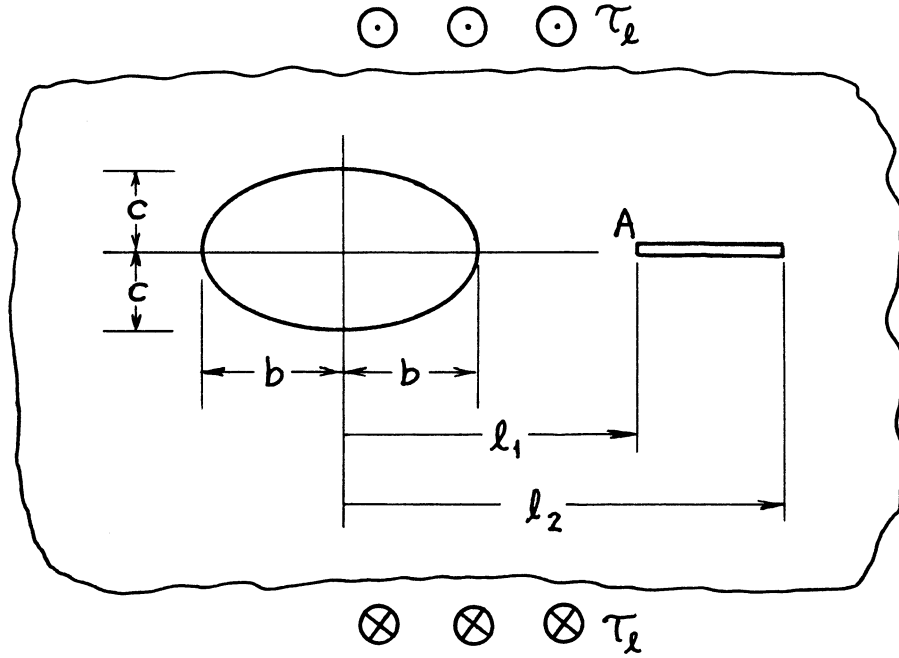
NOTE: The values of F_B for $a/(d-R) \rightarrow 1$ were taken from p. 19.2 where $\lambda = 1$.



Method: Expansions of Complex Stress Potentials

Accuracy: Curves were drawn based on the results having 0.1% accuracy.

Reference: **Isida 1970a**



$$K_{III,A} = \tau_l \sqrt{\pi(b+c)} \frac{s_1 \left[-\frac{1}{2} \left(s_1 - \frac{1}{s_1} \right) \left(s_1 - \frac{1}{s_1} + s_2 - \frac{1}{s_2} \right) + \left(s_1 s_2 + \frac{1}{s_1 s_2} - 2 \right) \frac{E(k)}{K(k)} \right]}{\sqrt{\left(s_1^2 - \frac{b-c}{b+c} \right) (s_2 - s_1) \left(s_1 - \frac{1}{s_1} \right) \left(s_1 - \frac{1}{s_2} \right)}}$$

where

$$s_i = \frac{1}{b+c} \left[\ell_i + \sqrt{\ell_i^2 - (b^2 - c^2)} \right], \quad i = 1, 2$$

$$k = \frac{s_2 - s_1}{s_1 s_2 - 1}$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} d\varphi$$

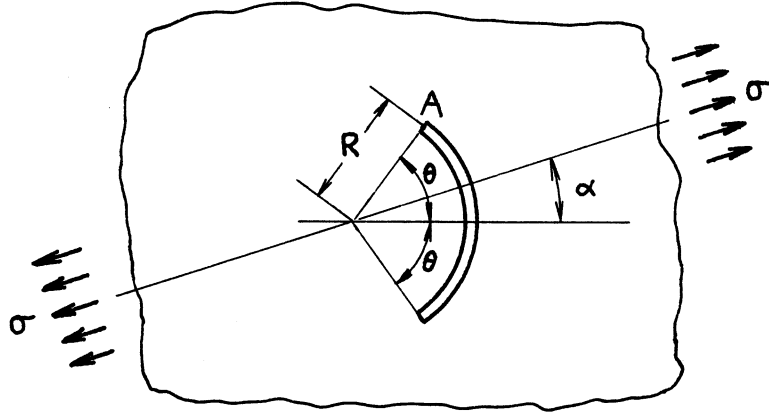
For Circular Hole,

$$b = c = R : \quad s_1 = \frac{\ell_1}{R}, \quad s_2 = \frac{\ell_2}{R}$$

Method: Complex Potentials

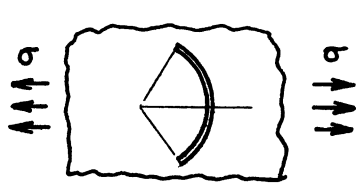
Accuracy: Exact

Reference: Yokobori 1972 (or Kamei 1974)

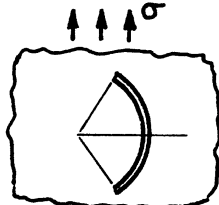


$$K_{IA} = \frac{\sigma\sqrt{\pi R \sin \theta}}{2} \left[\cos \frac{\theta}{2} \cdot \frac{1 - \cos 2\alpha \left(\sin \frac{\theta}{2} \right)^2 \left(\cos \frac{\theta}{2} \right)^2}{1 + \left(\sin \frac{\theta}{2} \right)^2} + \sin 2\alpha \left(\sin \frac{\theta}{2} \right)^3 + \cos \left(2\alpha - \frac{3\theta}{2} \right) \right]$$

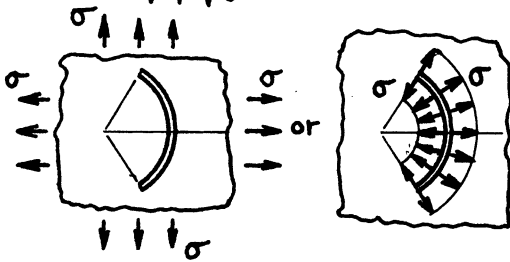
$$K_{IIA} = \frac{\sigma\sqrt{\pi R \sin \theta}}{2} \left[\sin \frac{\theta}{2} \cdot \frac{1 - \cos 2\alpha \left(\sin \frac{\theta}{2} \right)^2 \left(\cos \frac{\theta}{2} \right)^2}{1 + \left(\sin \frac{\theta}{2} \right)^2} - \sin 2\alpha \left(\sin \frac{\theta}{2} \right)^2 \cos \frac{\theta}{2} - \sin \left(2\alpha - \frac{3\theta}{2} \right) \right]$$



$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{\sigma\sqrt{\pi R \sin \theta}}{2 \left\{ 1 + \left(\sin \frac{\theta}{2} \right)^2 \right\}} \cdot \begin{Bmatrix} \cos \frac{\theta}{2} \left[2 - 4 \left(\sin \frac{\theta}{2} \right)^2 - 3 \left(\sin \frac{\theta}{2} \right)^4 \right] \\ \sin \frac{\theta}{2} \left[4 - 2 \left(\sin \frac{\theta}{2} \right)^2 - 3 \left(\sin \frac{\theta}{2} \right)^4 \right] \end{Bmatrix}$$



$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{\sigma\sqrt{\pi R \sin \theta}}{2 \left\{ 1 + \left(\sin \frac{\theta}{2} \right)^2 \right\}} \cdot \begin{Bmatrix} \cos \frac{\theta}{2} \left(\sin \frac{\theta}{2} \right)^2 \left[4 + 3 \left(\sin \frac{\theta}{2} \right)^2 \right] \\ \sin \frac{\theta}{2} \left[-2 + 2 \left(\sin \frac{\theta}{2} \right)^2 + 3 \left(\sin \frac{\theta}{2} \right)^4 \right] \end{Bmatrix}$$

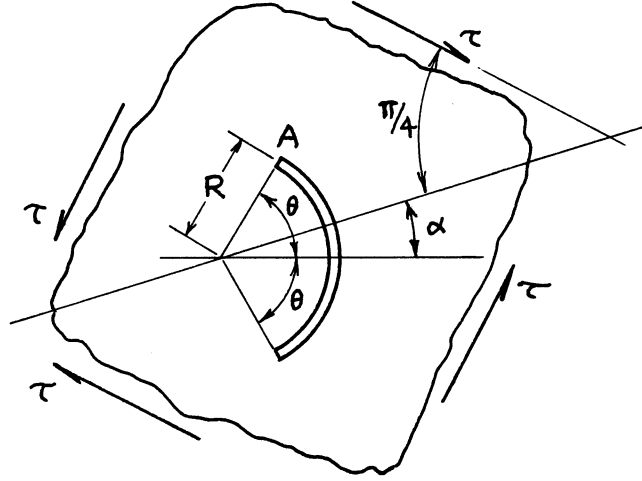


$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{\sigma\sqrt{\pi R \sin \theta}}{1 + \left(\sin \frac{\theta}{2} \right)^2} \begin{Bmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{Bmatrix}$$

Method: Conformal Mapping (Muskhelishvili)

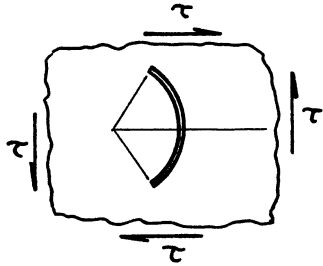
Accuracy: Exact

References: Muskhelishvili 1933; Sih 1962b

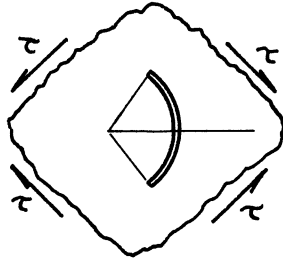


$$K_{IA} = \tau \sqrt{\pi R \sin \theta} \left[\frac{-\cos 2\alpha \left(\sin \frac{\theta}{2} \right)^2 \left(\cos \frac{\theta}{2} \right)^3}{1 + \left(\sin \frac{\theta}{2} \right)^2} + \sin 2\alpha \left(\sin \frac{\theta}{2} \right)^3 + \cos \left(2\alpha - \frac{3\theta}{2} \right) \right]$$

$$K_{IIA} = \tau \sqrt{\pi R \sin \theta} \left[\frac{-\cos 2\alpha \left(\sin \frac{\theta}{2} \right)^3 \left(\cos \frac{\theta}{2} \right)^2}{1 + \left(\sin \frac{\theta}{2} \right)^2} - \sin 2\alpha \left(\sin \frac{\theta}{2} \right)^2 \cos \frac{\theta}{2} - \sin \left(2\alpha - \frac{3\theta}{2} \right) \right]$$



$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \tau \sqrt{\pi R \sin \theta} \begin{Bmatrix} 3 \sin \frac{\theta}{2} \left(\cos \frac{\theta}{2} \right)^2 \\ \cos \frac{\theta}{2} \left[3 \left(\sin \frac{\theta}{2} \right)^2 - 1 \right] \end{Bmatrix}$$

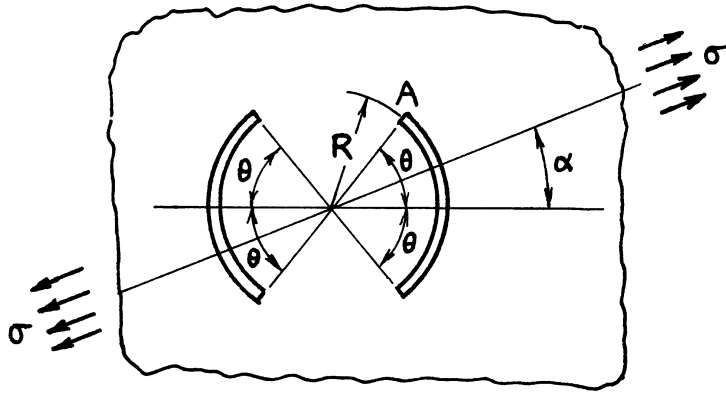


$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{\tau \sqrt{\pi R \sin \theta}}{1 + \left(\sin \frac{\theta}{2} \right)^2} \begin{Bmatrix} \cos \frac{\theta}{2} \left[1 - 4 \left(\sin \frac{\theta}{2} \right)^2 - 3 \left(\sin \frac{\theta}{2} \right)^4 \right] \\ \sin \frac{\theta}{2} \left[3 - 2 \left(\sin \frac{\theta}{2} \right)^2 - 3 \left(\sin \frac{\theta}{2} \right)^4 \right] \end{Bmatrix}$$

Method: Superposition of **page 21.1**

Accuracy: Exact

Reference: **Tada 1985**



$$H = H(\theta) = E(k)/K(k)$$

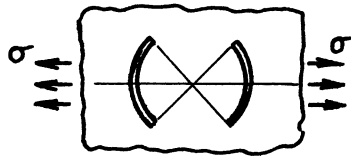
$$k = \sin \theta$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

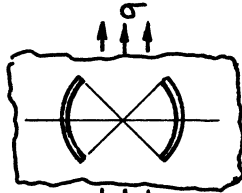
$$K_{IA} = \frac{\sigma}{2} \sqrt{\pi R \tan \theta} \left[\frac{H}{2-H} + \cos 2\alpha \left\{ 2 + \cos^2 \theta - \frac{2(1 + \sin^2 \theta)}{2-H} \right\} + \sin 2\alpha \cot \theta \left(1 + \sin^2 \theta - \frac{\cos^2 \theta}{H} \right) \right]$$

$$K_{IIA} = \frac{\sigma}{2} \sqrt{\pi R \cot \theta} \left[\frac{1-H}{2-H} - \cos 2\alpha \left(\cos^2 \theta - \frac{1 + \sin^2 \theta}{2-H} \right) - \sin 2\alpha \sin \theta \cos \theta \right]$$



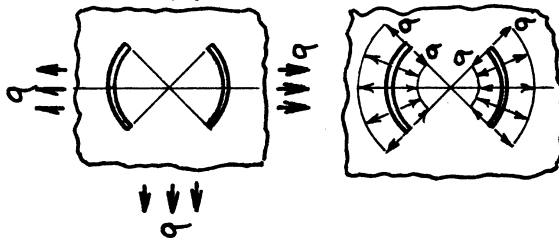
$$K_I = \frac{\sigma}{2} \sqrt{\pi R \tan \theta} \left(1 + \cos^2 \theta - \frac{2 \sin^2 \theta}{2-H} \right)$$

$$K_{II} = \frac{\sigma}{2} \sqrt{\pi R \cot \theta} \left(\frac{3-H}{2-H} \right) \sin^2 \theta$$



$$K_I = \frac{\sigma}{2} \sqrt{\pi R \tan \theta} \left[-3 - \cos^2 \theta + \frac{2(2 + \sin^2 \theta)}{2-H} \right]$$

$$K_{II} = \frac{\sigma}{2} \sqrt{\pi R \cot \theta} \left[1 + \cos^2 \theta - \frac{2 + \sin^2 \theta}{2-H} \right]$$



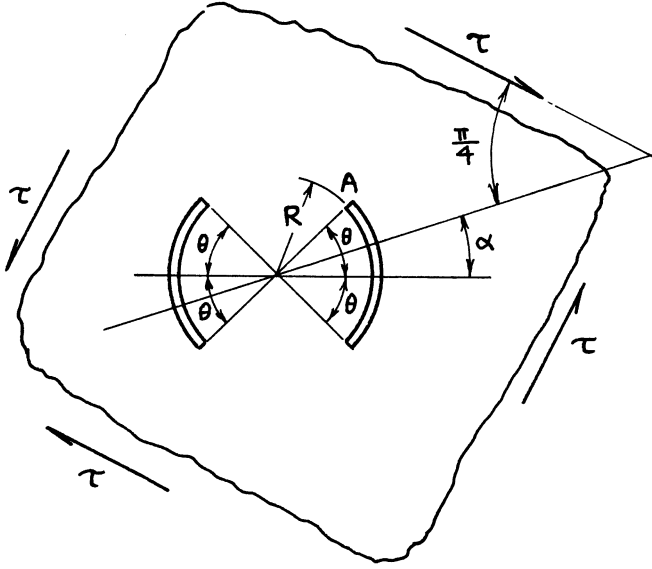
$$K_I = \sigma \sqrt{\pi R \tan \theta} \left(\frac{H}{2-H} \right)$$

$$K_{II} = \sigma \sqrt{\pi R \cot \theta} \left(\frac{1-H}{2-H} \right)$$

Method: Conformal Mapping (Muskhelishvili)

Accuracy: Exact

Reference: Tada 1985



$$H = H(\theta) = E(k)/K(k)$$

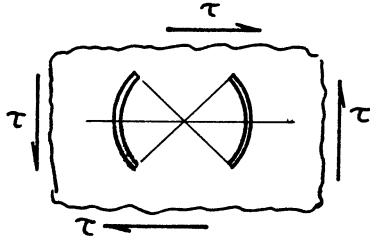
$$k = \sin \theta$$

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

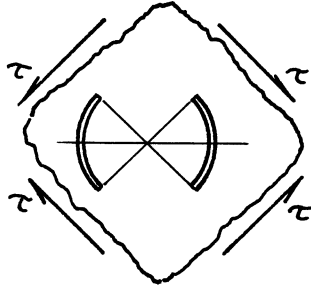
$$K_{IA} = \tau \sqrt{\pi R \tan \theta} \left[\cos 2\alpha \left\{ 2 + \cos^2 \theta - \frac{2(1 + \sin^2 \theta)}{2 - H} \right\} + \sin 2\alpha \cot \theta \left(1 + \sin^2 \theta - \frac{\cos^2 \theta}{H} \right) \right]$$

$$K_{IIA} = \tau \sqrt{\pi R \cot \theta} \left[-\cos 2\alpha \left(\cos^2 \theta - \frac{1 + \sin^2 \theta}{2 - H} \right) - \sin 2\alpha \sin \theta \cos \theta \right]$$



$$K_I = \tau \sqrt{\pi R \cot \theta} \left(1 + \sin^2 \theta - \frac{\cos^2 \theta}{H} \right)$$

$$K_{II} = \tau \sqrt{\pi R \cot \theta} (-\sin \theta \cos \theta)$$



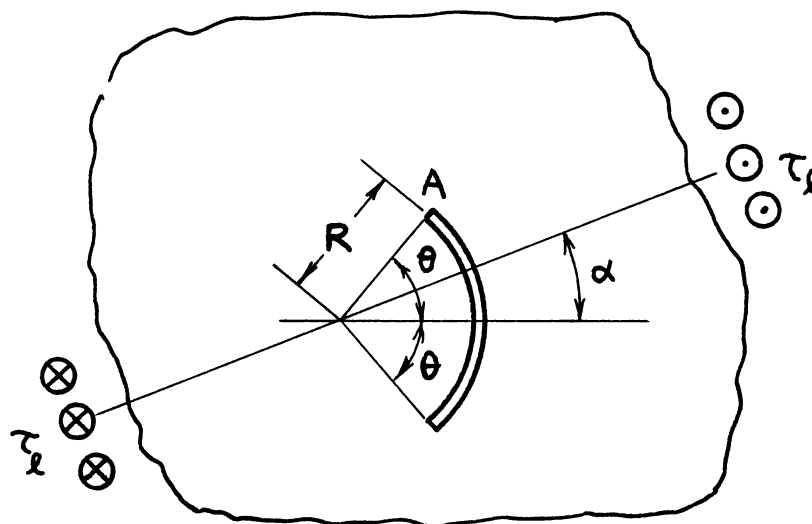
$$K_I = \tau \sqrt{\pi R \tan \theta} \left[2 + \cos^2 \theta - \frac{2(1 + \sin^2 \theta)}{2 - H} \right]$$

$$K_{II} = \tau \sqrt{\pi R \cot \theta} \left[-\cos^2 \theta + \frac{1 + \sin^2 \theta}{2 - H} \right]$$

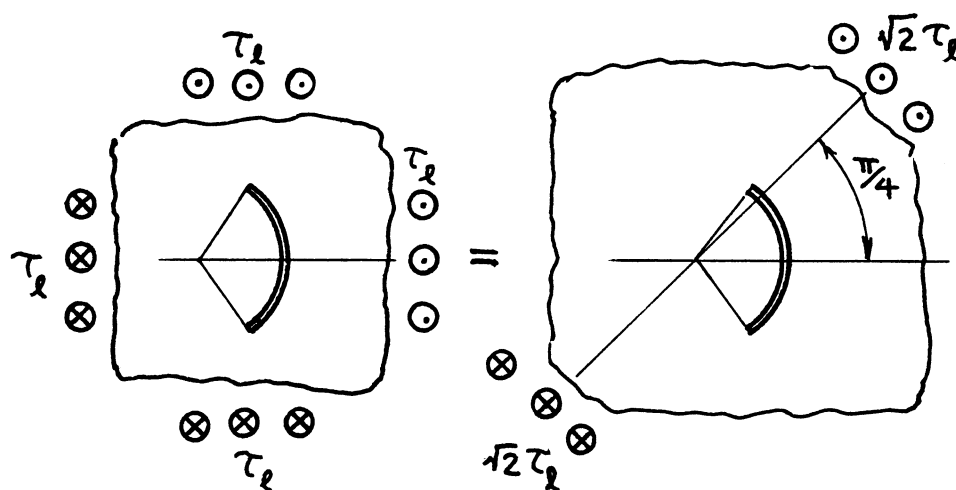
Method: Superposition of **page 21.3**

Accuracy: Exact

Reference: **Tada 1985**



$$K_{III A} = \tau_\ell \sqrt{\pi R \sin \theta} \cos\left(\alpha - \frac{\theta}{2}\right)$$

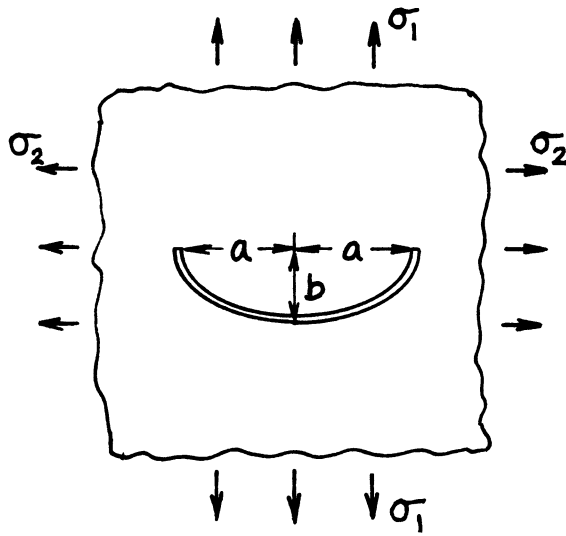


$$\begin{aligned} K_{III} &= \tau_\ell \sqrt{\pi R \sin \theta} \left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2} \right) \\ &= \sqrt{2} \tau_\ell \sqrt{\pi R \sin \theta} \cos\left(\frac{\pi}{4} - \frac{\theta}{2}\right) \end{aligned}$$

Method: Conformal Mapping

Accuracy: Exact

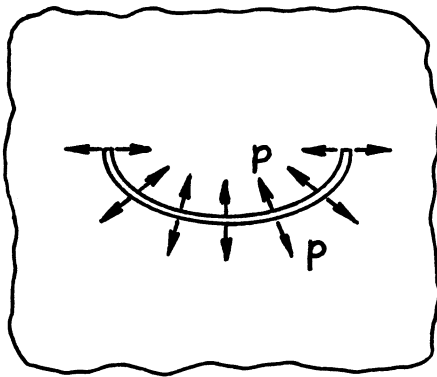
References: Sih 1965a



Semi-Elliptical Arc Crack

$$K_I = \sqrt{\pi a} \left[\sigma_1 F_{I1} \left(\frac{b}{a} \right) + \sigma_2 F_{I2} \left(\frac{b}{a} \right) \right]$$

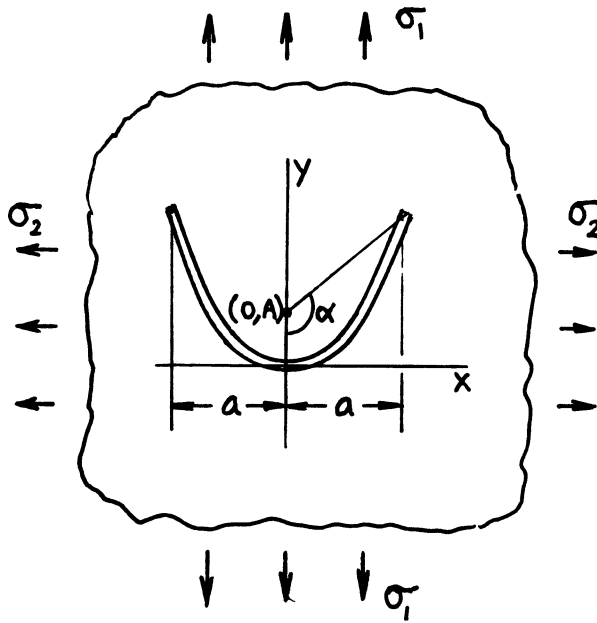
$$K_{II} = \sqrt{\pi a} \left[\sigma_1 F_{II1} \left(\frac{b}{a} \right) + \sigma_2 F_{II2} \left(\frac{b}{a} \right) \right]$$



$$K_I = p\sqrt{\pi a} \left[F_{I1} \left(\frac{b}{a} \right) + F_{I2} \left(\frac{b}{a} \right) \right]$$

$$K_{II} = p\sqrt{\pi a} \left[F_{II1} \left(\frac{b}{a} \right) + F_{II2} \left(\frac{b}{a} \right) \right]$$

See Narendran 1982.



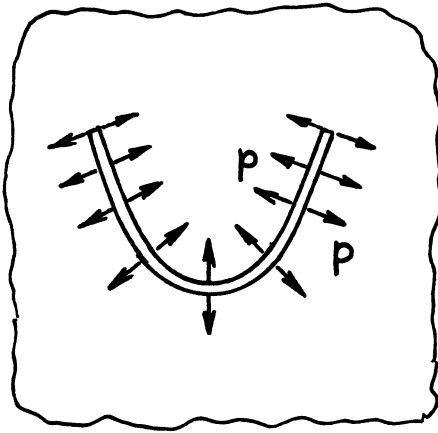
Parabolic Arc Crack:

$$x^2 = 4Ay, \quad |x| \leq a$$

$$\alpha = 2 \tan^{-1} \frac{a}{2A}$$

$$K_I = \sqrt{\pi a} [\sigma_1 F_{I1}(\alpha) + \sigma_2 F_{I2}(\alpha)]$$

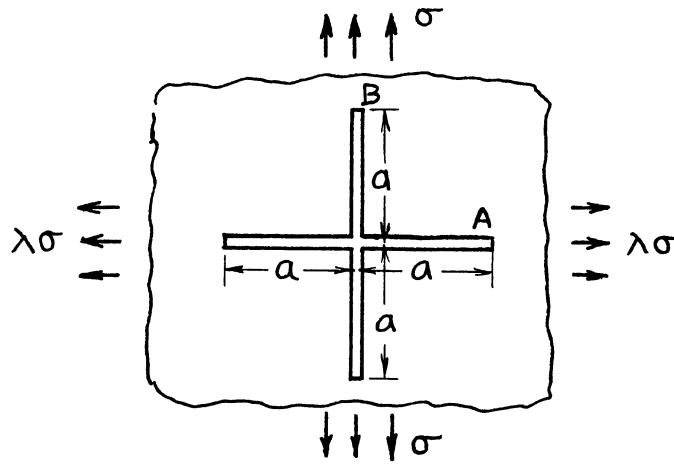
$$K_{II} = \sqrt{\pi a} [\sigma_1 F_{II1}(\alpha) + \sigma_2 F_{II2}(\alpha)]$$



$$K_I = p\sqrt{\pi a} [F_{I1}(\alpha) + F_{I2}(\alpha)]$$

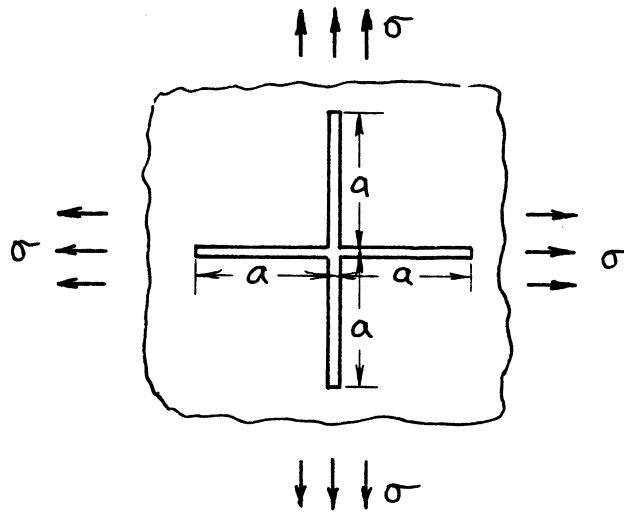
$$K_{II} = p\sqrt{\pi a} [F_{II1}(\alpha) + F_{II2}(\alpha)]$$

See Narendran 1982.



$$K_{IA} = \sigma\sqrt{\pi a} (1.0863 - .2227\lambda)$$

$$K_{IB} = \sigma\sqrt{\pi a} (-.2227 + 1.0863\lambda)$$

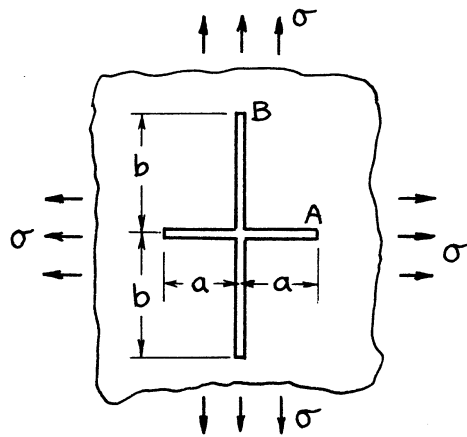


$$K_I = \sigma\sqrt{\pi a} (.8636)$$

Method: A Special Case of **page 21.10** and/or **page 21.9**

Accuracy: 0.1%

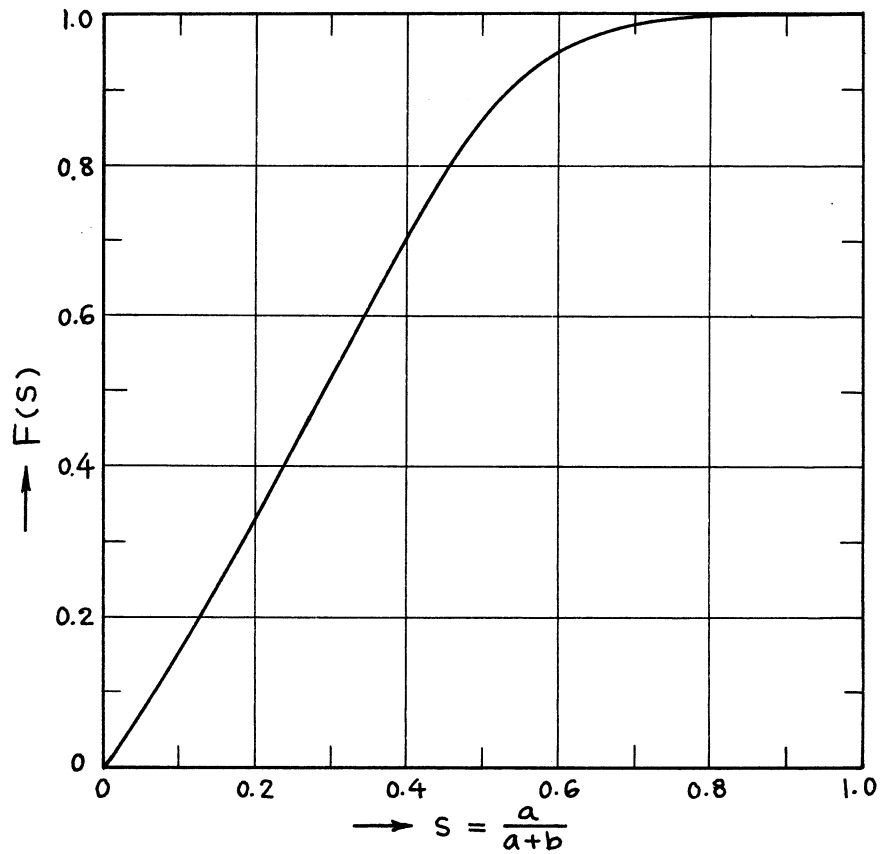
References: **Tada 1985**; also **Stallybrass 1969**; **Rooke 1969**



$$K_{IA} = \sigma\sqrt{\pi a} \cdot F(s)$$

$$K_{IB} = \sigma\sqrt{\pi b} \cdot F(1-s)$$

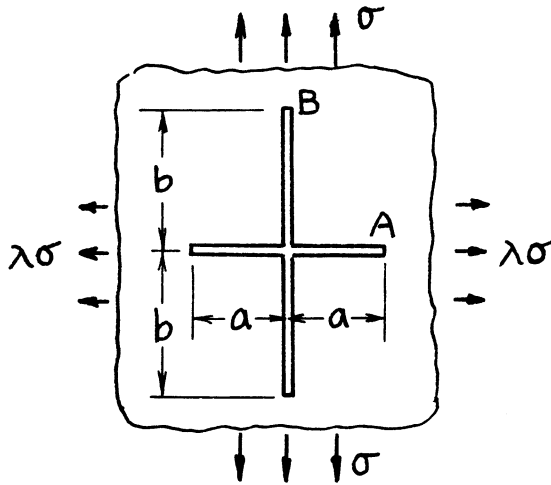
$$s = \frac{a}{a+b}$$



Method: Alternating Method (Special Case of **Page 21.10**; $\lambda = 1$)

Accuracy: Curve is based on numerical values with 0.1% accuracy.

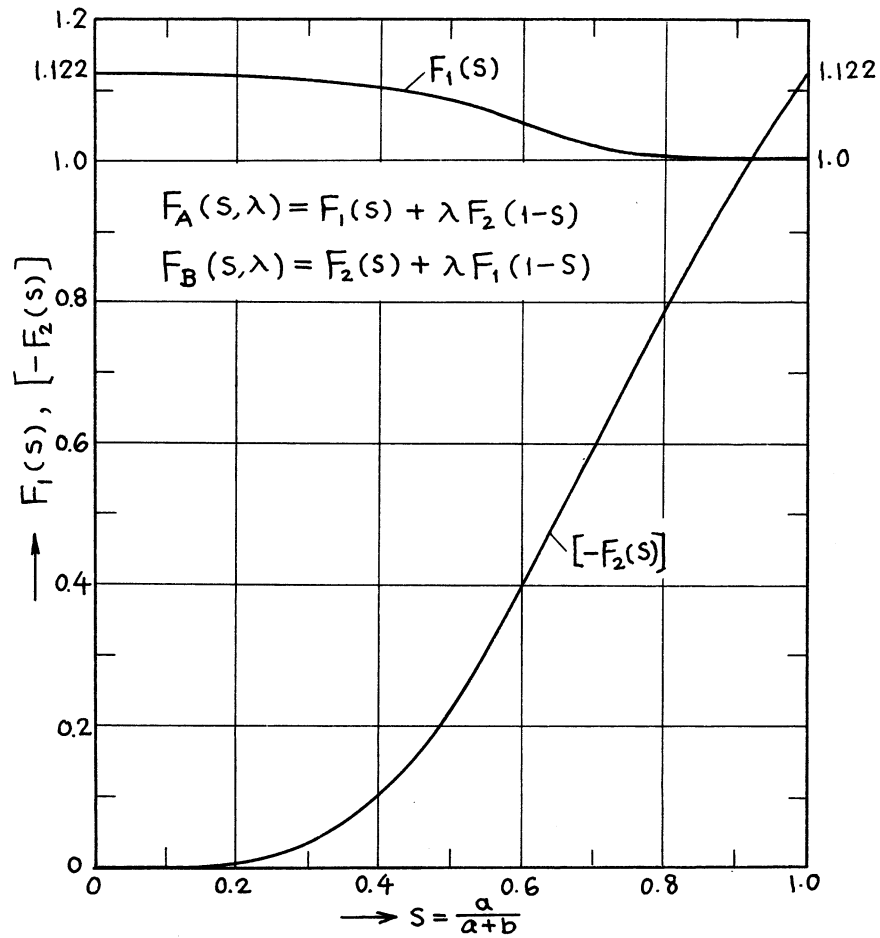
Reference: **Tada 1985**



$$K_{IA} = \sigma \sqrt{\pi a} \cdot F_A(s, \lambda)$$

$$K_{IB} = \sigma \sqrt{\pi b} \cdot F_B(s, \lambda)$$

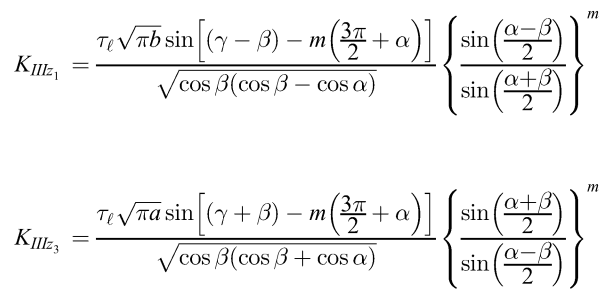
$$s = \frac{a}{a+b}$$



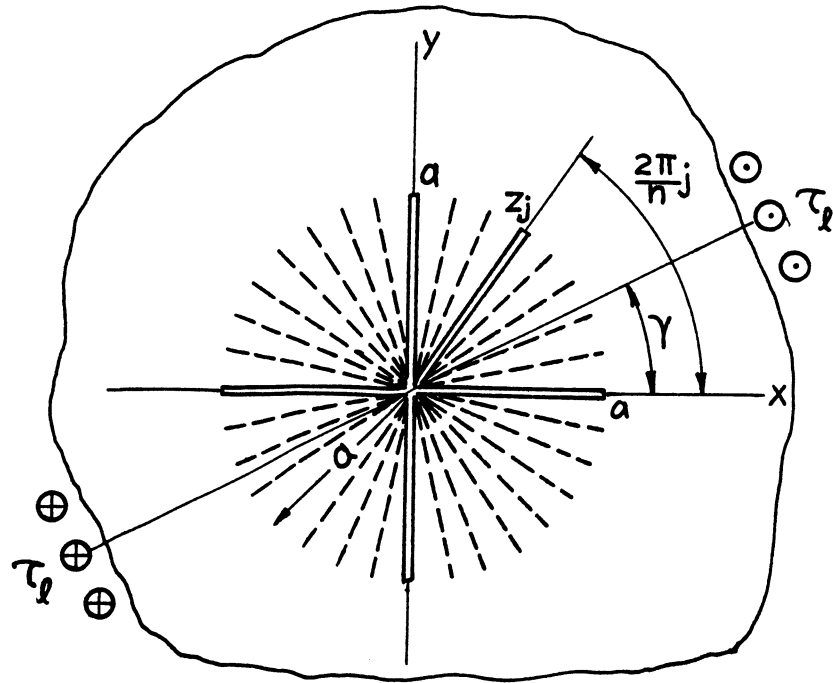
Method: Alternating Method (Simultaneous Integral Equations)

Accuracy: Curves are based on numerical values with 0.1% accuracy.

Reference: **Tada 1985**


$$\sin \beta = m \sin \alpha$$

Method: Conformal Mapping
Accuracy: Exact
Reference: **Sih 1965a**



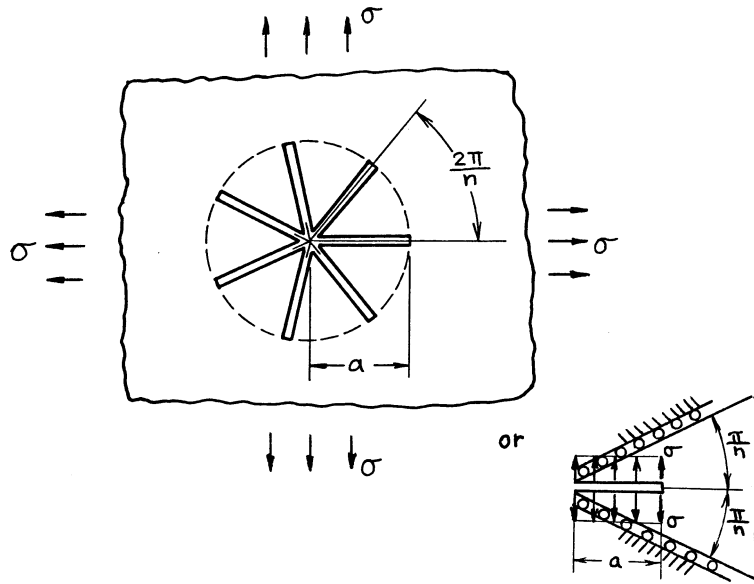
$$K_{IIIz_j} = (-1)^{1+j} 2\sqrt{2} \cdot 2^{-2/n} \tau_\ell \sqrt{\pi a} \frac{1}{\sqrt{n}} \sin\left(\gamma + \frac{2\pi}{n}j\right)$$

$$n = 1, 2, 3, \dots$$

Method: Conformal Mapping

Accuracy: Exact

Reference: Sih 1965a



$$K_I = \sigma \sqrt{\pi a} \cdot F(n)$$

$$F(n) = \frac{1}{\sqrt{n}} G(n)$$

$$G(n) = 2\sqrt{1 - \frac{1}{n}}, \quad n \geq 2 \quad (\text{Ouchterlony})$$

$$\text{or} \quad G(n) = 2 - 1.050 \frac{1}{n} - .243 \frac{1}{n^2}, \quad n \geq 1 \quad (\text{Tada})$$

$$n \rightarrow \infty : F(n) \rightarrow \frac{2}{\sqrt{n}}$$

Total Area of Crack Opening:

$$A = \frac{\sigma \pi a^2}{E'} S(n)$$

$$S(n) = \{G(n)\}^2$$

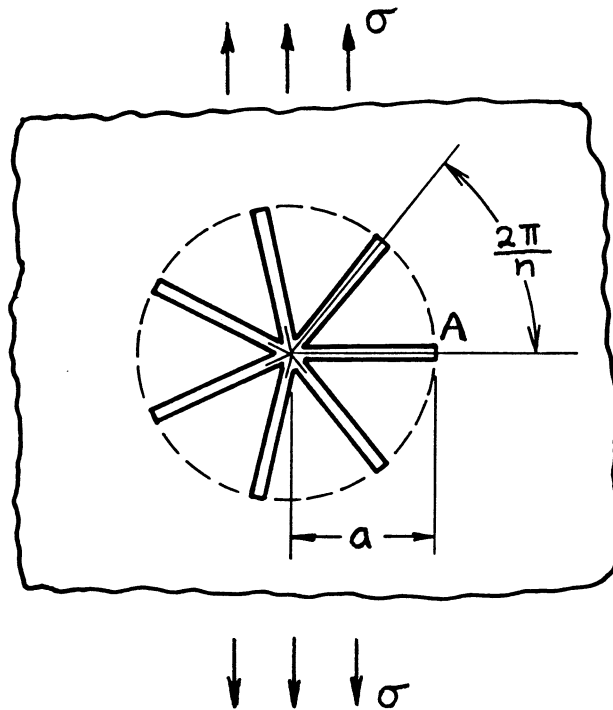
$$n \rightarrow \infty : S(n) \rightarrow 4$$

$$\text{or} \quad A_{n \rightarrow \infty} = 2A_{n=2}$$

Methods: Integral Transform and Integral Equation, Conformal mapping

Accuracy: Both formulas for K_I have better than 0.5% accuracy; A better than 1%

References: **Westmann 1964; Williams 1971; Kitagawa 1975; Ouchterlony 1975; Tada 1985**



$$(K_I)_{\max} = K_{IA} = \sigma \sqrt{\pi a} \cdot F_A(n)$$

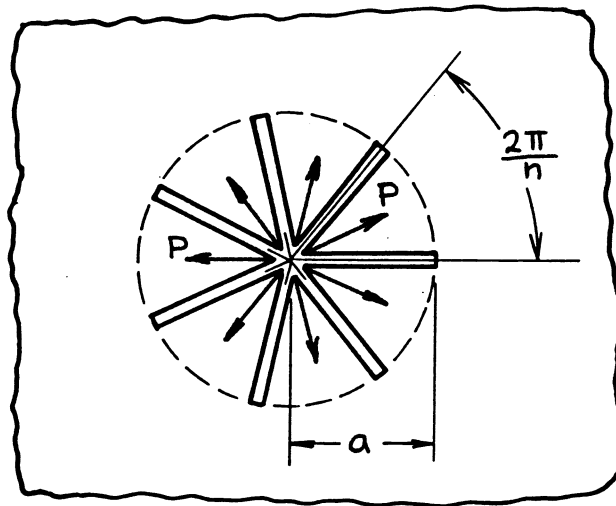
$$F_A(n) = \frac{1}{\sqrt{n}} \left(3 - 3.25 \frac{1}{n} - .64 \frac{1}{n^2} + 1.6 \frac{1}{n^3} \right) \quad n \geq 1$$

$$n \rightarrow \infty : F_A(n) = \frac{3}{\sqrt{n}}$$

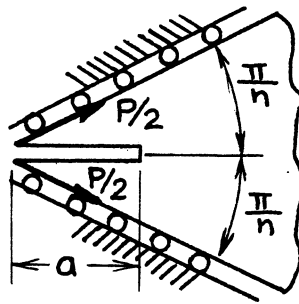
Method: Estimated by Interpolation (based on results for $n = 1, 2, 4,$ and ∞)

Accuracy: Expected to be within 1%

Reference: **Tada 1985**



or

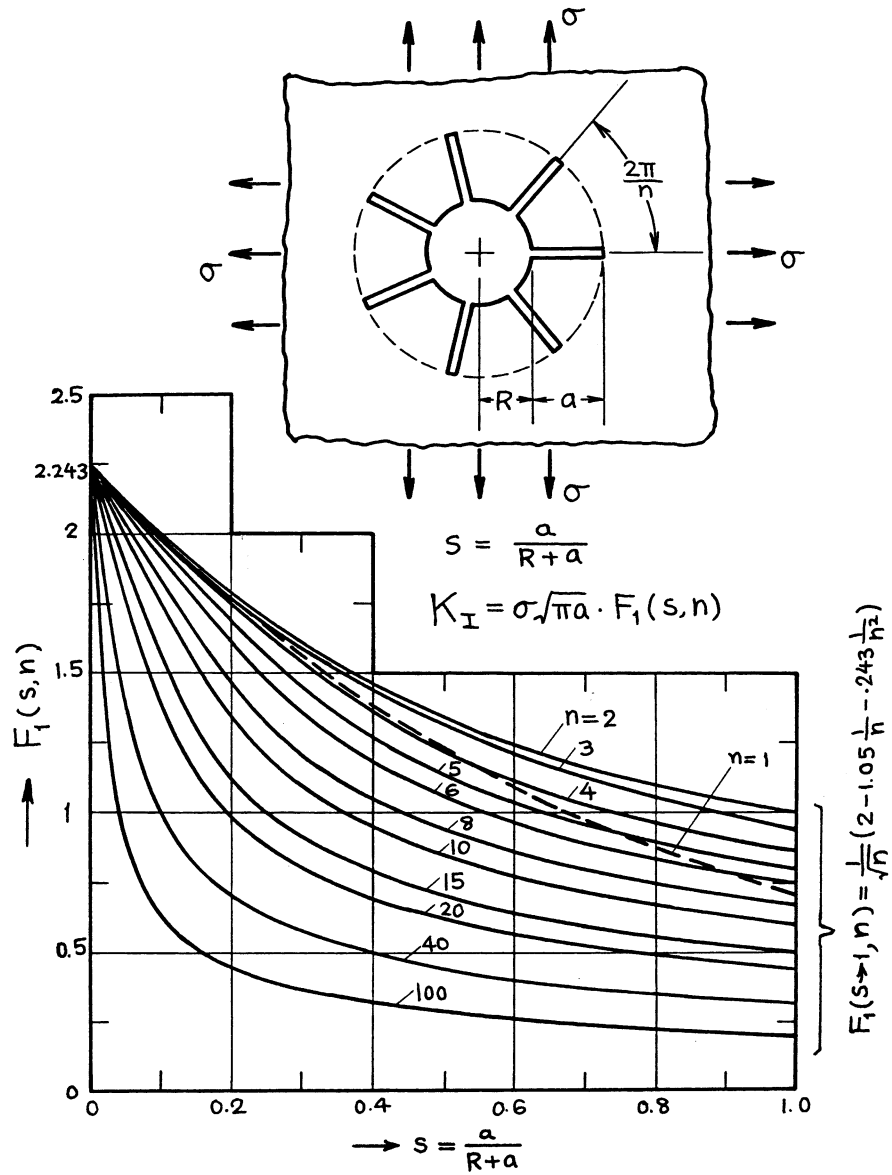
 $(n \geq 2)$ 

$$K_I = \frac{P}{\sqrt{a}} \cdot \frac{1}{\sqrt{\frac{2\pi}{n} + \sin \frac{2\pi}{n}}}$$

Method: Integral Transform

Accuracy: Exact

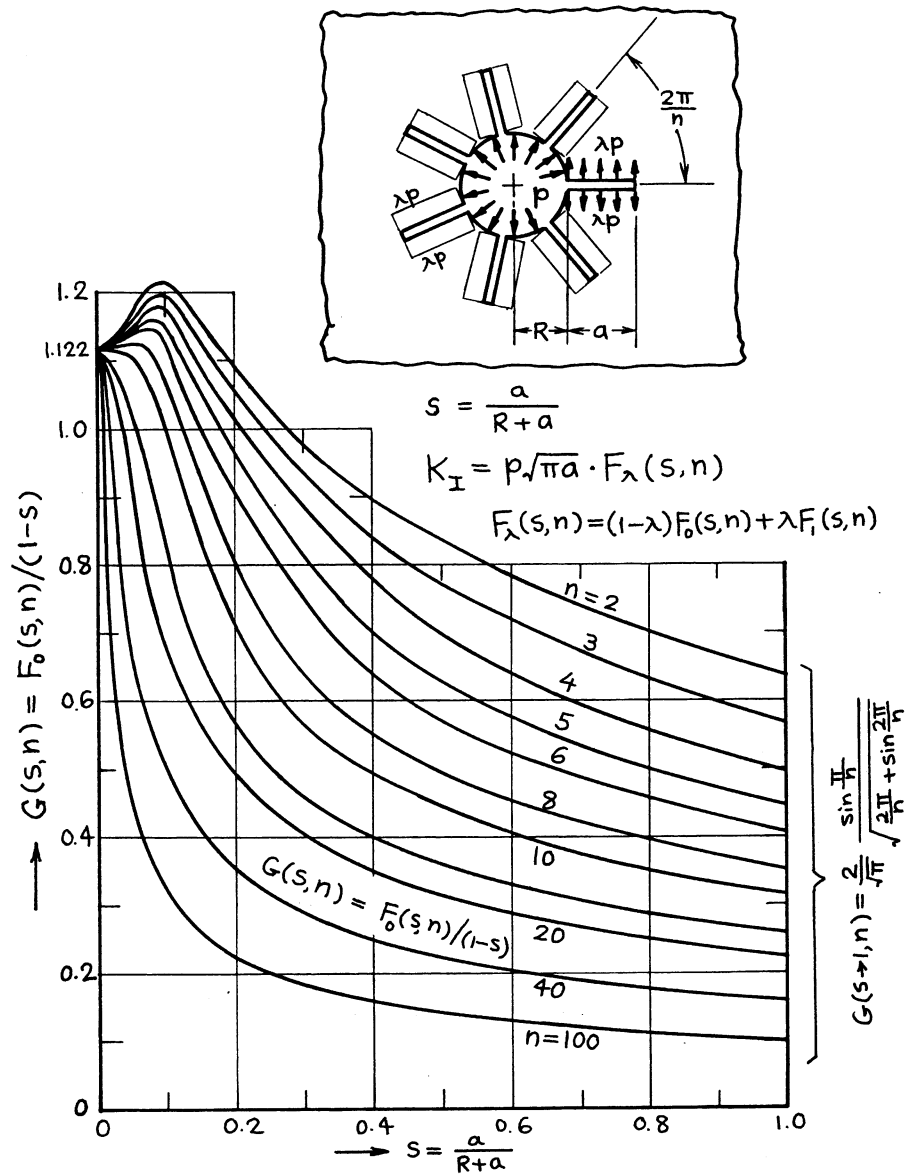
Reference: Ouchterlony 1975, 1976



Methods: Conformal Mapping (Ouchterlony; $s \leq 0.6, n \leq 15$), Asymptotic Interpolation (Tada; $s > 0.6, n \leq 15$ and $0 < s < 1, n > 15$)

Accuracy: 1%

References: Ouchterlony 1975; Tada 1985



Methods: Conformal Mapping (Ouchterlony; $s \leq 0.6$, $n \leq 15$), Asymptotic Interpolation (Tada; $s > 0.6$, $n \leq 15$ and $0 < s < 1$, $n > 15$)

Accuracy: 1%

References: **Ouchterlony 1975; Tada 1985**

NOTE: For $F_1(s, n)$, see page 21.16.